

AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

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ABSTRACT

Sparse autoencoders (SAEs) have become the standard instrument for extracting sparse, interpretable latents from transformer activations, and the quantitative metrics used to certify their quality now underwrite much of mechanistic interpretability practice. We subject these metrics to a decisive sanity check: we ask whether the SAE quality metrics and automatic explanation pipelines in common use can distinguish a trained transformer from a randomly initialized one whose parameters are drawn i.i.d. from a Gaussian. Evaluating the entire Pythia suite across several distinct randomization schemes, we establish that, in many settings—and, tellingly, most acutely at the largest scales—SAEs trained on randomly initialized transformers yield auto-interpretability scores and reconstruction metrics that are indistinguishable from those obtained on the corresponding trained models, while both remain sharply separated from a chance-level control. It follows that a high aggregate auto-interpretability score is not, on its own, evidence that an SAE has recovered learned, computationally relevant features: such scores may equally reflect statistical structure inherent in the data or inductive biases of the architecture that survive randomization. We therefore argue that common SAE metrics should be treated as useful but strictly insufficient proxies for mechanistic interpretability, and we make the case for two concrete correctives—the routine use of randomized-transformer baselines and targeted measures of feature ‘abstractness,’ the latter of which we show recovers precisely the trained-versus-random distinction that the aggregate metrics miss.

1 INTRODUCTION

Sparse autoencoders (SAEs) have become a central tool in mechanistic interpretability research, where they are used to disentangle the internal representations of neural networks by learning sparse, interpretable features from network activations (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). The construction is by now canonical: an autoencoder with a high-dimensional hidden layer is trained to reconstruct activations subject to a sparsity constraint (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024), with the aim of recovering the underlying concepts, or ‘features,’ that the network has learned (Park et al., 2023; Wattenberg and Viégas, 2024). Because the space of such architectures is large and the features they recover cannot be inspected by hand at scale, progress in SAE design rests squarely on quantitative evaluation metrics—chief among them auto-interpretability scores, which quantify the agreement between automatically generated explanations and observed activation patterns (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a). The validity of these metrics as proxies for mechanistic understanding is thus a load-bearing assumption of the field, and it is precisely this assumption that we interrogate.

Any interpretability method that purports to characterize the computation a model acquires through training must, at a minimum, yield evaluation metrics that separate features learned during training from artifacts of the data distribution or the model architecture. The canonical means of establishing this property is to compare a method’s output on a trained model against a strong null model—most naturally, one with randomly initialized weights (Adebayo et al., 2020). We apply exactly this sanity check to SAEs and demonstrate that several of the most widely used quantitative metrics fail to distinguish the trained setting from the randomized one. Concretely, SAEs trained on transformers with random parameters recover latents whose auto-interpretability scores (Bills et al., 2023; Paulo

et al., 2024) are strikingly close to those obtained from a fully trained model—so close, across the Pythia suite, that the aggregate scores cannot be used to tell the two apart.

This finding compels a reconsideration of what SAE quality metrics actually certify. A high auto-interpretability score, taken in isolation, does not establish that an SAE has identified complex, learned computation; it is consistent with a far more mundane explanation. Such a score may instead reflect simpler statistical properties of the training data (Dooms and Wilhelm, 2024) or architectural inductive biases that are already present prior to any training. A randomly initialized network, after all, still performs a rudimentary form of computation—preserving or even amplifying the sparse structure of its inputs (Section 4)—so that from this vantage point an SAE may be faithfully interpreting a simple but genuine, architecture-induced computation rather than a learned one.

We emphasize that this is not a claim that SAE features are uninformative: some features obtained from trained models plainly originate in learned computation. The precise point is that the aggregate metrics in common use are, on their own, inadequate for deciding whether a given SAE has captured such complex features. These results carry direct and actionable implications for mechanistic interpretability practice. They establish the need for more discriminating metrics that separate architectural and data-driven artifacts from genuinely learned computation, and they make the case that interpretability techniques should, as a matter of routine, be validated against appropriately constructed null models.

Finally, we advance a mechanistic account of why these patterns arise, framed as two complementary hypotheses: (1) the input data already exhibits superposition, which randomly initialized neural networks largely preserve; and (2) randomly initialized neural networks amplify, or even introduce, superposed structure in the input data (for example, given dense input generated i.i.d. from a Gaussian). In Section 4 we develop a sequence of toy models that establish the plausibility of both hypotheses, deferring to future work only the narrower question of which mechanism predominates for the randomized transformers of the main text.

2 RELATED WORK

Sparse dictionary learning Superposition, under a different name, is one of the foundational observations of computational neuroscience. Olshausen and Field (1996; 1997) established that the receptive fields of simple cells in the mammalian visual cortex arise from sparse coding: the representation of a comparatively large number of signals (sensory information) through the simultaneous activation of a small number of elements (neurons). Coding theory, in turn, provides the natural framing for the effort to recover the ‘underlying signals’ responsible for neural network activations (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) approximates a set of input vectors by linear combinations of a comparatively small number of learned basis vectors, where the learned basis is typically overcomplete—of dimension greater than that of the inputs. Established SDL algorithms include Independent Component Analysis (ICA), which recovers a linear representation of the data whose components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) constitute a neural-network instantiation of this idea (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014): an autoencoder with a single hidden layer many times wider than the input activation vectors is trained under an objective that imposes or incentivizes sparsity in its hidden-layer activations, thereby recovering this structure. Throughout, we refer to a single neuron (dimension) in the autoencoder’s hidden layer as a **latent**.

Mechanistic interpretability It has become standard to construe ‘features,’ or concepts, in language models as low-dimensional subspaces of the internal model activations (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). Wherever such sparse, ‘superposed’ structure exists, one expects to be able to ‘intervene on’ or ‘steer’ the activations—to modify or replace them so as to express different concepts and thereby influence model behavior (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs are the predominant approach for discovering such features: one typically trains a single autoencoder to reconstruct the activations of a single network layer, for example the transformer residual stream (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). A range of SAE

architectures has been proposed, differing primarily in the activation function applied after the linear encoder (Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024). SAEs have further been trained under alternative objectives (Braun et al., 2024; Farnik et al., 2025) and applied to several layers simultaneously (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

Beyond reconstruction error and preservation of the underlying model’s performance, SAEs have been evaluated on whether they capture specific concepts (Gurnee et al., 2023; Gao et al., 2024) or factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), and on whether their features support the targeted ‘unlearning’ of concepts (Karvonen et al., 2024b).

Automatic neuron description Because SAEs routinely learn tens of thousands of latents, describing each by hand is infeasible, and the field has converged on automated description. Yun et al. (2021) identify the tokens in a text dataset that maximally activate a dictionary element and inspect the resulting activation patterns manually. The now-dominant alternative is to collect a latent’s activation patterns over a text dataset and prompt a large language model to explain them (e.g. Bills et al., 2023; Foote et al., 2023). These methods have been adopted extensively across the literature (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024).

Bills et al. (2023) generate an explanation for the activation patterns of a language-model neuron over dataset examples, simulate those patterns from the explanation, and score the explanation by the agreement between observed and simulated activations—a procedure now widely known as auto-interpretability (in the sense of self-interpreting). Paulo et al. (2024) introduce classification-based measures of the fidelity of automatic descriptions that are inexpensive to compute relative to full activation simulation, together with an open-source pipeline that implements them. Choi et al. (2024) apply best-of- k sampling to generate multiple explanations, each conditioned on a different subset of the examples that maximally activate a neuron, and, notably, fine-tune Llama-3.1-8B-Instruct on the top-scoring explanations to obtain inexpensive ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) observed that neurons may become polysemantic incidentally. A polysemantic neuron (basis dimension) of a network layer represents multiple interpretable concepts (Elhage et al., 2022; Scherlis et al., 2023); it is therefore unsurprising that individual neurons in a randomly initialized network can already be polysemantic. The phenomenon we study, by contrast, is *superposition* (Elhage et al., 2022; Chan, 2024)—a property of the representation carried across an entire network layer rather than of any individual neuron. Superposition is precisely what allows a layer as a whole to represent more (sparse) features than it has (dense) neurons, through a sparse code in which only a few concepts are active at any given token position. This distinction is central to our argument: it is superposition, not incidental single-neuron polysemanticity, that an SAE is meant to disentangle.

Training only the embeddings Zhong and Andreas (2024) showed that transformers acquire surprising algorithmic capabilities when the embeddings alone are trained and all other parameters are held fixed, demonstrating that the behavior of a randomly initialized transformer can be shaped to a considerable extent by training only a small subset of its parameters. Our setting is fundamentally different: beyond our reliance on SAEs, we randomize *all* parameters, embeddings included, in our ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants. The ‘Re-randomized excl. embeddings’ variant does retain the pre-trained embeddings, but—crucially, and in direct contrast to that setting—we do not train them: we freeze the pre-trained embeddings and randomize every other weight (Section 3).

Random transformers for board games Karvonen et al. (2024c) found that SAEs extracted meaningful structure from chess games considerably more effectively for pre-trained transformers than for randomly weighted ones. Board-game data, however, differs from language data in a respect that we expect to be decisive here. Language is plausibly sparse—a concept such as ‘serendipitous’ occurs only rarely—and, critically, this sparse structure is ‘aligned’ with conceptual meaning. No analogous alignment holds for board games: a useful concept such as a knight fork need not occur sparsely across positions. This contrast explains why the trained-versus-random gap reported for board games need not carry over to language, and it is language that we study.

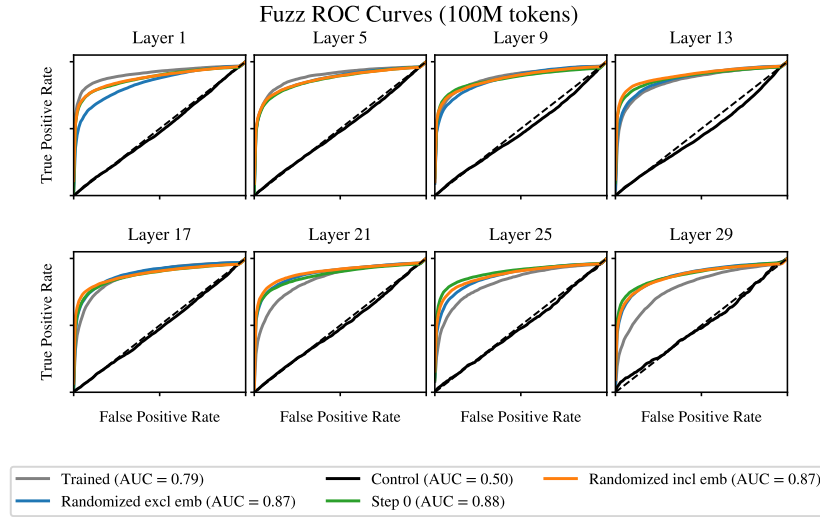


Figure 1: Layerwise ‘fuzzing’ ROC curves for Pythia-6.9b, each computed over 100 latents sampled uniformly per SAE. The trained model (gray) and the three randomized variants (colored) overlap closely at every layer, while the control (black) stays near chance (dotted diagonal). Because the trained and randomized curves are mutually indistinguishable by AUROC, this figure directly demonstrates that aggregate AUROC cannot, on its own, attribute a set of latents to learned computation. Figure 2 reports the remaining metrics and model sizes, and Appendix E confirms that the overlap persists across multiple random seeds.

Random one-layer transformers Bricken et al. (2023) found that auto-interpretability scores discriminated effectively between random and trained one-layer transformers. Our results are fully consistent with this and, moreover, explain when such discrimination should be expected: auto-interpretability scores for randomized models are relatively low for smaller models (e.g., Pythia-70m), but the gap between randomized and trained models narrows substantially at larger scales (e.g., Pythia-6.9b), where the metrics cease to distinguish them.

3 RESULTS

We trained per-layer SAEs on the residual-stream activation vectors of transformer language models drawn from the Pythia suite, spanning 70M to 7B parameters (Biderman et al., 2023). Covering nearly two orders of magnitude in scale allows us to characterize the trained-versus-randomized comparison as an explicit function of model size, rather than at a single, potentially unrepresentative operating point. For each underlying transformer, we compared SAEs trained on the following five variants:

- **Trained:** the standard, trained model, serving as the reference point.
- **Re-randomized incl. embeddings:** every model parameter, embeddings included, is re-initialized by sampling i.i.d. Gaussian noise whose mean and variance match those of the corresponding original, trained weight matrix, thereby preserving each matrix’s first two moments while destroying all learned structure.
- **Re-randomized excl. embeddings:** as above, except that the embedding and unembedding matrices are left untouched—identical to those of the original, trained model—so as to isolate the effect of randomizing the transformer body while retaining a learned token representation.
- **Step-0:** for Pythia models, the step0 revisions provide the original weights at initialization, prior to any learning (Biderman et al., 2023), and thus constitute a randomization drawn from the model’s own initialization scheme.

- **Control:** the original, trained model, but with each input token embedding replaced at inference time by an i.i.d. standard Gaussian sample drawn afresh at every occurrence, so that no token retains a consistent embedding vector. By construction this variant is a chance-level reference: auto-interpretability should perform no better than chance on it, and it therefore calibrates the lower bound against which every other variant is read.

In our primary experiments, we trained SAEs on 100M tokens from the RedPajama dataset (Weber et al., 2024) with an activation buffer of 10M tokens; Appendix C reproduces a subset of these experiments at one billion tokens and obtains materially identical results, ruling out limited training data as an explanation for our findings. To manage compute at scale, we adopted a layer-sampling schedule that grows coarser with model size: for models with fewer than 410M parameters we trained an SAE at every layer; for Pythia-1b at every second layer; and for Pythia-6.9b at every fourth layer.

Unless stated otherwise, we trained k -sparse autoencoders (TopK SAEs; Makhzani and Frey 2014; Gao et al. 2024) with expansion factor $R = 64$ and sparsity $k = 32$. Our conclusions are robust to these hyperparameters: sweeping SAEs trained on Pythia-160m across expansion factors at powers of two from 16 to 128 and sparsities of 16 and 32 leaves them unchanged (Figure 18). Our training implementation follows Belrose et al. (2025), and our evaluations follow Caden et al. (2025) and Karvonen et al. (2024a), so that the metrics we report are exactly those in standard use rather than measures tailored to our setting.

Auto-interpretability The auto-interpretability of a feature explanation is assessed by supplying the explanation, which names a concept, to a classifier tasked with predicting whether that concept is present in the text inputs, and then scoring the classifier by a standard metric such as the area under the receiver operating characteristic (ROC) curve (AUROC). Paulo et al. (2024) formalize two such classification tasks for evaluating feature explanations, ‘fuzzing’ and ‘detection.’ In ‘fuzzing’ scoring, positive and negative token examples for a given latent (those with non-zero and zero activation, respectively) are delimited with special characters, and a language model is prompted to identify, from the latent’s explanation alone, which examples have been correctly delimited. In ‘detection,’ the language model is instead asked to identify which examples contain activating tokens for the feature. Bills et al. (2023) originally proposed ‘simulation’ scoring, based on the correlation between predicted and observed activations, but that procedure is comparatively expensive to compute.

Except where noted, we adopt ‘fuzzing’ as our primary measure of auto-interpretability, because it has been shown to correlate with the more expensive simulation scoring (Paulo et al., 2024) while remaining tractable at our scale; the corresponding AUROC curves for ‘detection’ scoring, which yield the same conclusions, are provided in Appendix B. For each trained SAE—that is, each combination of underlying model, variant, and layer—we sampled 100 latents uniformly at random and computed their auto-interpretability scores, following the implementation of Paulo et al. (2024). Both explanations and predictions were generated with Meta-Llama-3.1-70B-Instruct-AWQ-INT4, which is both larger than the 8B models used by Choi et al. (2024) and, unlike Bills et al. 2023, fully open-source—so that our pipeline is at once stronger and more reproducible than these precedents.

The auto-interpretability scores of the trained and randomized models are far closer to one another than either is to the control (Figures 1 and 2). This near-coincidence of the trained and randomized ROC scores establishes directly that ‘fuzzing’ auto-interpretability, applied to SAE latent explanations in isolation, does not meaningfully distinguish these underlying models.

Evaluation We evaluated the standard suite of SAE metrics alongside the auto-interpretability AUROC across Pythia models from 70M to 6.9B parameters. The pattern established for auto-interpretability generalizes to this broader battery of metrics: the randomly initialized and re-randomized variants (Figure 2; blue, green, orange) resemble the trained model (gray) far more closely than they resemble the control (black).

The separation between the control and the remaining variants is pronounced and consistent: both the cosine similarity between the original activation and its SAE reconstruction and the explained variance are markedly lower for the control, and its reconstruction error grows across layers while that of every other variant declines. This behavior admits a principled explanation. A Gaussian is the maximum-entropy distribution at fixed mean and variance (Jaynes, 2003), so the control’s Gaussian activations are in this sense the ‘least structured’ inputs and correspondingly the hardest for an SAE to reconstruct. As these Gaussian-distributed activations propagate through successive

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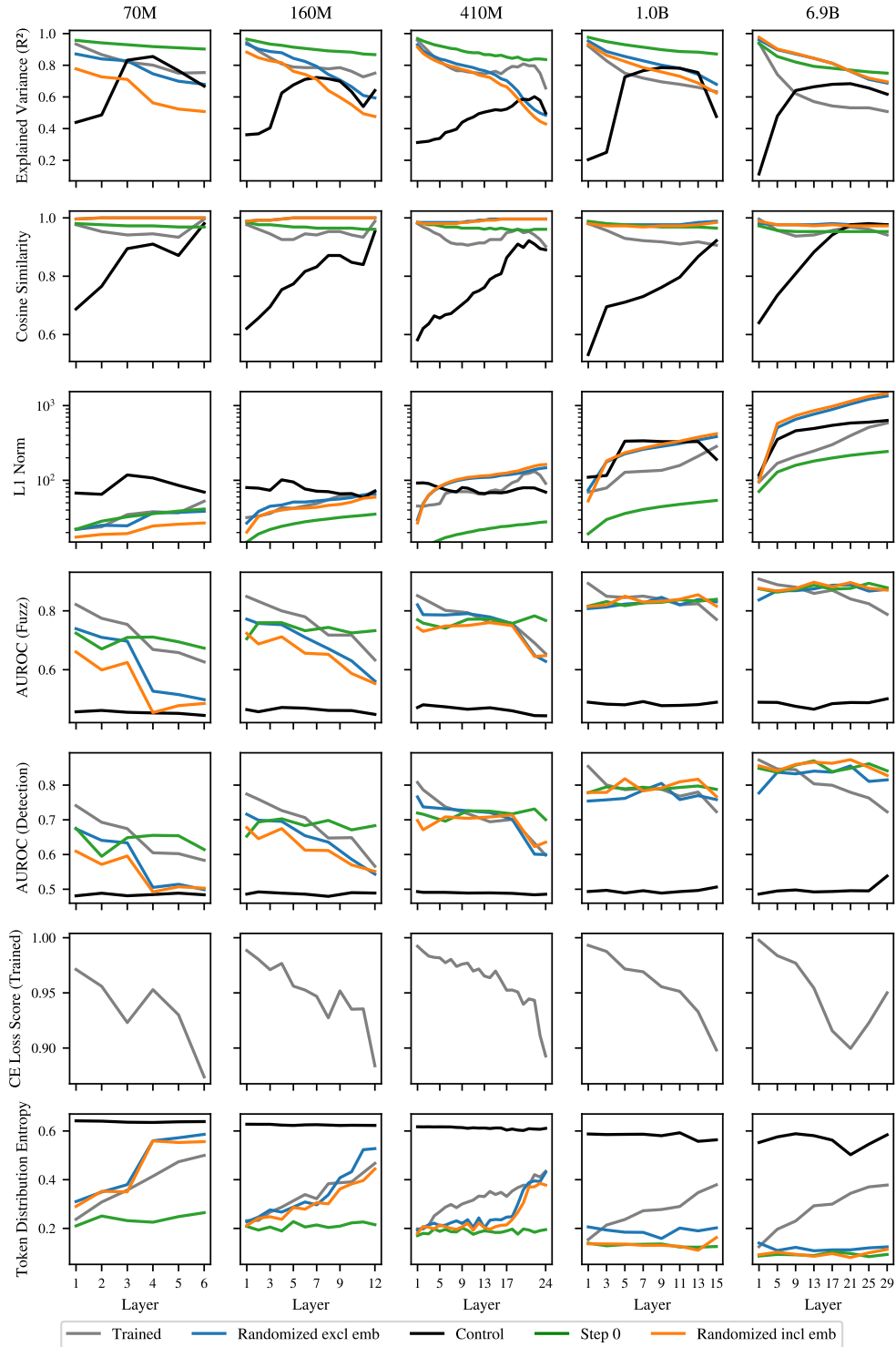


Figure 2: Sparse-autoencoder evaluation metrics across the Pythia suite (70M to 6.9B parameters), plotted by layer, with model size along the columns and metric along the rows. Across every scale, the trained and randomized variants track one another closely—most tightly for the larger models—while only the control departs systematically from the rest. The central pattern this figure documents is unambiguous: despite their fundamentally different initializations, every variant except the control attains comparable performance on the standard metric suite. This holds jointly across reconstruction cosine similarity, explained variance, L^1 norm, CE loss score, and auto-interpretability AUROC, so the trained-versus-random collapse is not specific to any single metric.

layers, we expect them to become less Gaussian and more ‘sparse,’ and hence progressively easier to reconstruct (Section 4)—an expectation borne out by the observed decline in reconstruction error for the non-control variants.

Among the non-control variants, the re-randomized models (blue and orange) resemble the trained model more closely than does the variant at initialization (green), an effect most visible in the L^1 norm at larger scales. We attribute this to the treatment of parameter norms: whereas a trained model’s parameter norms can diverge substantially from their values at initialization, our re-randomization procedure was designed precisely to preserve the trained model’s parameter norms. The per-layer parameter scale plausibly matters—for instance, in governing the growth of activations along the residual stream (Liu et al., 2020). The AUROC plots reveal a further regularity that reinforces our main claim: for every variant except the control, AUROC increases with model size. We interpret this as features becoming more specific as SAE size grows: in smaller SAEs each latent must account for more of the input, so the classification task becomes easier for the larger SAEs associated with larger models—and, consequently, the trained and randomized variants become *harder* to tell apart at exactly the scales that matter most in practice.

Row five of Figure 2 plots the cross-entropy (CE) loss score, or loss recovered, against layer: the increase in loss when the original activations are replaced by their SAE reconstructions, normalized by the increase when they are instead replaced by zeros (‘ablated’). For the trained variant, our SAEs perform on par with comparable results in the literature (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024), confirming that they are of standard quality and that the trained-versus-random similarity we report does not stem from weak SAEs. This metric is, however, interpretable only for the trained variant: for every randomized variant the model loss is poor regardless of whether the original or the reconstructed activations are used, so the score does not carry its usual meaning—a caveat we state explicitly to forestall over-reading the CE panel for the randomized variants.

Latent explanation complexity The coincidence of the aggregate auto-interpretability scores and evaluation metrics does not, however, imply that the underlying features are the same. We hypothesized that SAEs applied to trained versus randomized transformers recover qualitatively different features even where their aggregate scores agree. Specifically, we expected the randomized variants to yield relatively simple features keyed to surface characteristics of the input text, in contrast to the more complex, abstract features characteristic of trained transformers (Templeton et al., 2024). This qualitative gap is directly visible in the recovered features: Appendix J presents a random sample of features together with their maximally activating dataset examples for each variant of Pythia-6.9b, and Appendix L provides more detailed feature dashboards.

A substantial fraction of SAE latents activate on only a single token, or a small set of distinct tokens, across a text dataset (e.g., Lin and Bloom, 2024; Doods and Wilhelm, 2024). This observation motivates a simple but discriminating complexity measure: given a latent’s set of maximally activating examples, we ask whether it fires on a single token ID or on many distinct IDs. We operationalize this by the entropy of the observed distribution of the latent’s activation over tokens, where higher entropy indicates more ‘spread out,’ less token-specific activation. We take this distribution to be the total latent activation per token, aggregated over the set of maximally activating examples used to generate explanations in the auto-interpretability pipeline—so that the measure is computed on exactly the examples that drive the explanations themselves. We show the relationship between entropy and ‘fuzzing’ AUROC score for individual latents in Appendix H.

The last row of Figure 2 reports the entropy of these token-ID activation distributions, and the three variant classes separate cleanly—in sharp contrast to the auto-interpretability scores, which do not. The control exhibits uniformly high entropy, exactly as expected: because each token’s embedding is resampled i.i.d. from a Gaussian on every occurrence, no token retains a consistent embedding vector (Section 3), so activation cannot concentrate on particular IDs. The trained variant shows entropy that rises across layers—the deeper the layer, the less its latents’ maximally activating examples concentrate on a single token—again as expected, since later layers should encode more abstract features that are less tied to individual token embeddings. The randomized variants, by contrast, exhibit lower entropy than either the trained or the control variant, indicating that their latents fire specifically at one or a few token IDs.

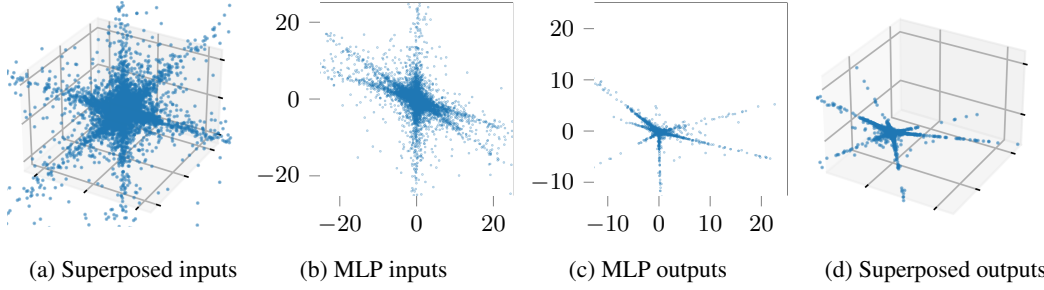


Figure 3: A worked illustration of the effect of a randomly initialized neural network on superposed input data. We draw 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$ and project them to $n_d = 2$ dense input features via an i.i.d. standard-normal matrix. The dense inputs are then passed through a two-layer ReLU MLP with hidden size $4n_d$, and $n_s = 3$ sparse outputs are recovered by inverting the projection matrix used above. The outputs remain superposed despite the intervening nonlinearity—and, indeed, appear *more* sharply superposed than the inputs (compare panels 3a and 3d), foreshadowing the amplification effect quantified in Section 4.2.

Taken together with the preceding results, this pinpoints a concrete dimension that the standard SAE quality and auto-interpretability metrics overlook entirely: the ‘abstractness’ of the recovered features. Token-distribution entropy is not, on its own, a direct measure of abstractness; but read alongside the near-identical auto-interpretability scores of the trained and randomized variants, it draws a clear and actionable distinction. The randomized variants recover simple, single-token features and do so about as ‘interpretably,’ by the aggregate scores, as the trained model; crucially, however, and unlike the trained variant, their features do not grow more complex with layer depth. Entropy thus recovers precisely the trained-versus-random separation that the aggregate metrics erase, and it does so at no additional modeling cost.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

In Section 1 we advanced two candidate explanations for the apparently high sparsity and interpretability of randomized transformers’ activations: either the input data already exhibits superposition that neural networks preserve, or neural networks themselves amplify—or even introduce—superposition into the input data. We examine both possibilities through a sequence of toy models that isolate each mechanism in turn. The evidence we obtain supports both mechanisms; what we defer to future work is only the finer question of which predominates for the randomized transformers of the main text.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We begin with a simplified generative model that makes precise the sense in which multiplication by a weight matrix W preserves superposition. Suppose superposed input data x is generated by first drawing n_s i.i.d. ‘sparse’ features z from a heavy-tailed Lomax distribution $z \sim \text{Lomax}(\alpha, \lambda)$. Projecting the higher-dimensional, sparse z down to the lower-dimensional, dense x through a matrix D and adding small-variance Gaussian noise Σ yields $x \sim \mathcal{N}(x; Dz, \Sigma)$. The decisive observation is that left-multiplication of x by an arbitrary matrix W leaves this structure intact: $x' = Wx$ is *itself* superposed, generated by the same model as x but with a transformed mapping WD from z to x and a correspondingly transformed noise covariance, $x' \sim \mathcal{N}(z; WDz, W\Sigma W^T)$. Superposition is therefore invariant under an arbitrary linear map, and in particular under the random weight matrices of an untrained network.

This invariance extends beyond the linear case to networks with nonlinearities, as the effect of passing the dense activations through a simple feed-forward network (MLP) makes visible. In the example of Figure 3, $n_s = 3$ sparse features are projected down to $n_d = 2$ dense features, and the MLP outputs remain superposed despite the intervening nonlinearity. The figure moreover suggests that networks may *amplify* superposition rather than merely preserve it: relative to the inputs (Figures 3a and 3b),

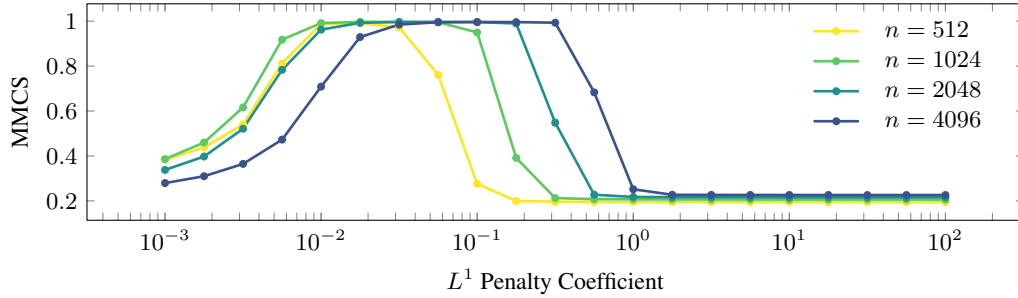


Figure 4: Mean max cosine similarity (MMCS) between the features learned by a standard SAE (its decoder weight vectors) and the data-generating features, as a function of the L^1 penalty coefficient in the training loss, following Sharkey et al. (2022). A ‘Goldilocks zone’ of penalty coefficients emerges within which the SAE recovers the data-generating features near-perfectly, provided it has enough latents to represent them—confirming that, in this controlled setting, the SAE recovers the true generative structure and thereby validating it as an instrument for the analysis that follows.

the outputs (Figures 3d and 3c) contain fewer points lying between the ‘arms’ of the distribution. We test this amplification hypothesis quantitatively next.

4.2 DO RANDOM NNs PRESERVE OR AMPLIFY SUPERPOSITION?

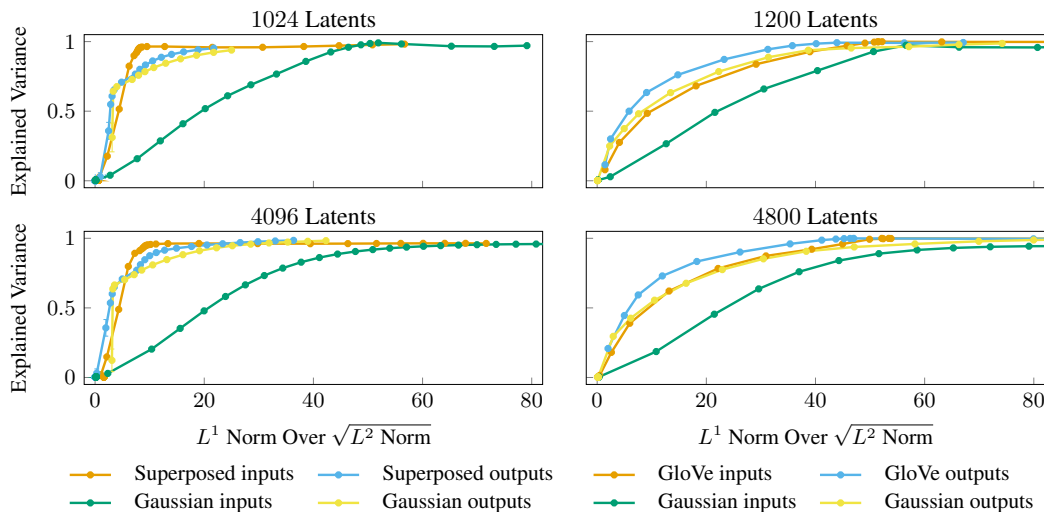
To test this suggestion quantitatively, we generated toy data following the procedure of Sharkey et al. (2022): ground-truth features are sampled on a hypersphere and assigned correlated coefficients such that only a small number are active at a time (Appendix I.1). We passed these inputs through a two-layer MLP at initialization and trained SAEs separately on the MLP inputs and on the MLP outputs, so that any change in recovered sparsity can be attributed to the network alone. As a control, we used Gaussian-distributed inputs whose mean and standard deviation matched those of the superposed toy data. Throughout, we used standard SAEs with an L^1 sparsity penalty (Appendix I.2).

Following Sharkey et al. (2022), we first confirmed that SAEs recover the ground-truth features underlying the data (Figure 4), as measured by the mean max cosine similarity (MMCS): for each data-generating feature we take its maximum cosine similarity with the SAE’s learned features (its decoder weight vectors), then average over data-generating features. The MMCS, however, is available only for the MLP inputs, where the data-generating features are known; analyzing the MLP outputs demands a different instrument. We therefore adopt the SAE’s ability to attain low reconstruction error at high sparsity as a proxy for the degree of superposition present in the training data, varying the L^1 penalty coefficient to trace Pareto frontiers of explained variance against sparsity (Figure 5a).

As expected, the SAEs achieved substantially greater sparsity at a given level of explained variance on the superposed inputs than on the Gaussian control (Figure 5a; orange and blue-green). The revealing comparison, however, is at the MLP outputs: the gap between the superposed outputs (the MLP outputs given superposed inputs) and the Gaussian outputs is far smaller, amounting to only slightly greater sparsity at a given level of explained variance. The outputs of a randomly initialized MLP thus attain a high level of sparsity that is largely insensitive to the input distribution—direct evidence that the network itself, rather than the input alone, supplies much of the observed sparsity, and hence that random networks amplify superposition rather than merely transmitting it. Additional sparsity measures and hyperparameter settings, all consistent with this conclusion, are reported in Appendix I.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If randomly initialized neural networks preserve or amplify superposition, then our main results (Section 3) may be attributable, at least in part, to superposition already present in the inputs to transformer language models. We test this hypothesis directly by applying the procedure of Section 4.2 to language data, training SAEs on pre-trained GloVe word vectors, on the embedding matrices of Pythia models, on the outputs of passing each through a randomly initialized two-layer MLP, and on



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Pareto frontiers of explained variance against sparsity, measured by the L^1 norm divided by the square root of the L^2 norm, for datasets that plausibly exhibit superposition, for Gaussian controls matched in mean and variance, and for the corresponding outputs obtained when each is passed through a randomly initialized two-layer MLP (Section 4.2). Each point corresponds to one choice of L^1 penalty coefficient. The effect is consistent across both the constructed toy data (left) and real GloVe word vectors (right): in every case, passing the inputs through the random MLP shifts the frontier toward greater sparsity, so the amplification of superposition by random networks holds for real linguistic representations and not only for synthetic data.

Gaussian controls. The setup is identical to that of Section 4.2, except that the number of data points is fixed by the number of word embeddings or tokens and we use a single random seed.

The gap between the Pareto frontiers of the GloVe word vectors and their matched Gaussian controls (Figure 5b) is smaller than the gap observed for the toy superposed datasets of Section 4.2 (Figure 5a), indicating that word embeddings are superposed, though less strongly than the deliberately constructed toy data. More tellingly, the Pareto frontiers for both input types again improve once the inputs are passed through a randomly initialized two-layer MLP. This replicates, now on real linguistic representations, the sparsification effect established on synthetic data, and it consolidates the central conclusion of this section: random networks ‘sparsify’ their inputs—that is, they increase the degree of apparent superposition—and both mechanisms hypothesized in Section 1 are operative in the settings we can measure.

5 LIMITATIONS

Our study establishes that auto-interpretability measures can return apparently meaningful, interpretable results for SAEs trained on randomly initialized models—models that are unlikely to harbor computationally interesting features. Exhaustive coverage of all datasets and architectures is infeasible, and we made a deliberate choice of test bed rather than an incidental one: we concentrated on the Pythia family—the de facto standard in mechanistic interpretability research (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024)—and on the RedPajama-V2 dataset, which is representative of the pre-training data used for contemporary language models and SAEs. Because these are precisely the models and data on which SAE evaluation is ordinarily conducted, the setting is a demanding and widely relevant test bed rather than a narrow one, and the finding bears directly on mainstream practice.

We generated explanations using the default model of the EleutherAI auto-interpretability framework (Caden et al., 2025); substituting alternative explainer models is a natural extension that could further illuminate aggregate behavior and explanation quality, though we have no reason to expect

it would overturn the effect, which is already stable across model scales, randomization schemes, hyperparameters, and training-data budgets. We are careful to delimit the claim precisely: we do *not* assert that SAEs extract no information from trained transformers beyond what they extract from randomized ones. Our claim is the sharper and fully supported one—that aggregate auto-interpretability measures do not, by themselves, certify the presence of interesting underlying features.

6 CONCLUSION

We applied sparse autoencoders to both trained and randomly initialized transformers and evaluated them with the standard suite of quantitative metrics. Our central empirical finding is decisive: under conditions that recur across the entire Pythia suite, these metrics—and aggregate auto-interpretability scores in particular—are strikingly similar in the two settings. Although features derived from trained transformers are qualitatively more complex and abstract, especially in later layers, the aggregate metrics routinely fail to register this distinction.

This result does not imply that SAEs trained on real models fail to learn meaningful computational features; rather, it exposes a concrete limitation in how the field currently evaluates them. A high aggregate auto-interpretability score is not, by itself, sufficient evidence for the discovery of complex, learned computation, since it may instead reflect simpler structure inherent in the data or the architecture that survives even under random weights. Our token-distribution-entropy analysis, though intended as a proof of concept, does more than diagnose the problem: it recovers precisely the differences in feature ‘abstractness’ that the aggregate auto-interpretability scores miss, demonstrating that a metric sensitive to learned computation is attainable and indicating the form such a metric might take. Developing more robust metrics that quantify the computational significance of the features SAEs discover is, accordingly, a priority for future work. Above all, our results establish the necessity of benchmarking interpretability techniques against strong, appropriately constructed null models such as the randomly initialized transformers used here; absent such baselines, discovered features cannot be confidently attributed to learning. We recommend that randomized-transformer controls become a standard component of SAE evaluation.

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A BROADER IMPACT

This work scrutinizes a method in active use for the mechanistic interpretability of LLMs and yields results that challenge prevailing assumptions about sparse autoencoders. By establishing that SAEs produce comparable aggregate auto-interpretability scores for random and trained transformers alike, our findings sharpen the question of what these SAE evaluation methods actually capture and, we hope, redirect effort toward metrics that measure it.

A clearer understanding of SAE quality metrics should support a better-informed search for improved SAE-like methods, and thereby help render these models more interpretable and mitigate potential harms. Because our contribution is a critical empirical study of a method already in widespread use—one that we show interprets random and trained transformers alike—we judge the risk of negative societal impact from this work to be minimal.

B AUTO-INTERPRETABILITY ROC CURVES

Figures 6, 8, and 12 document the close agreement between the ‘fuzzing’ AUROC of the trained and randomized SAEs for the 70M, 160M, and 1B models, and Figures 7, 9, and 13 establish the same agreement for the ‘detection’ AUROC across these model sizes. That the two scoring tasks yield the same conclusion confirms that our finding is a property of auto-interpretability scoring in general rather than an artifact of the particular ‘fuzzing’ variant emphasized in the main text.

B.1 PYTHIA 70M

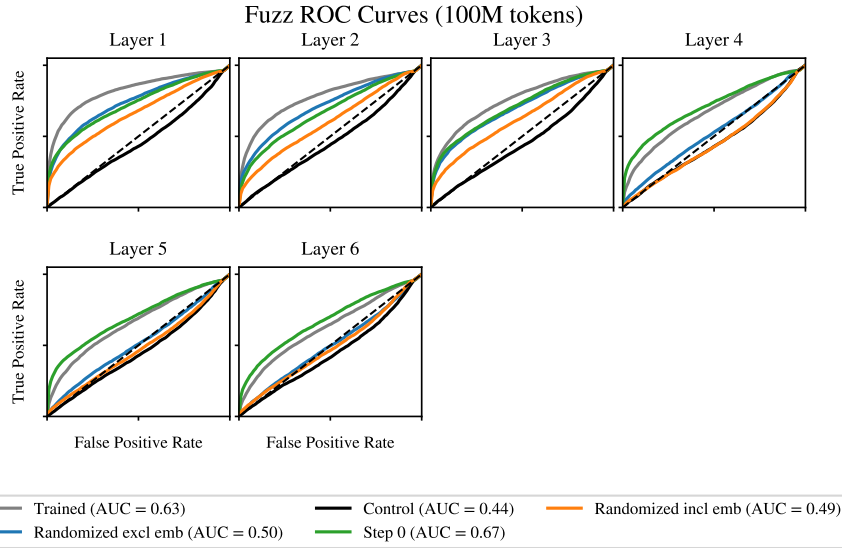


Figure 6: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-70m, computed over 100 SAE latents per layer. The SAE variants perform comparably to one another, and all degrade uniformly as the layer index increases—though, as noted in the main text, the trained–randomized gap is widest at this smallest scale.

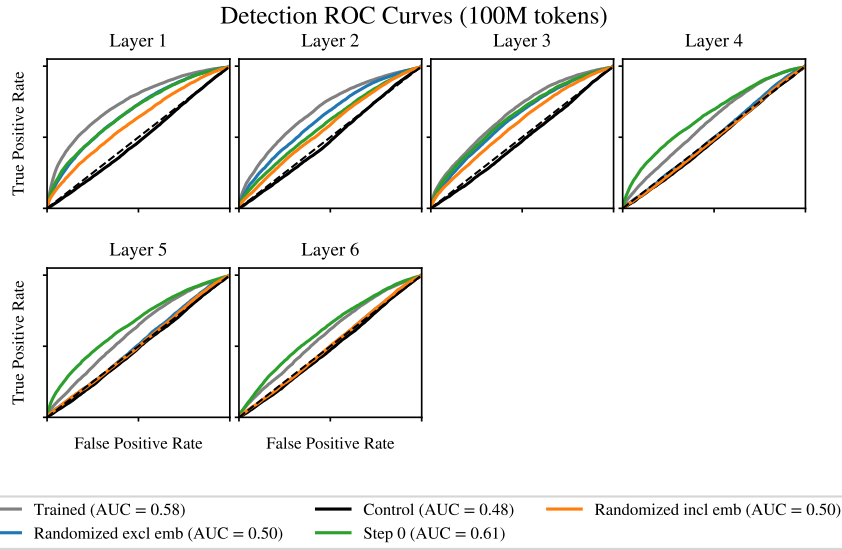


Figure 7: ‘Detection’ auto-interpretability ROC curves for Pythia-70m, computed over 100 SAE latents per layer. As under ‘fuzzing,’ the SAE variants perform comparably and degrade uniformly as the layer index increases, so the two scoring tasks agree.

B.2 PYTHIA 160M

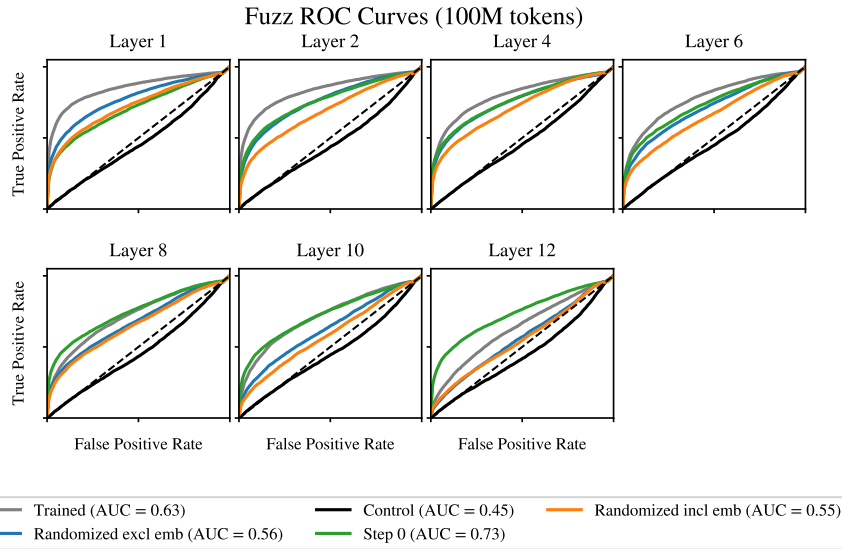


Figure 8: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-160m, computed over 100 SAE latents per layer. As at 70M, the SAE variants perform comparably to one another, degrading uniformly as the layer index increases.

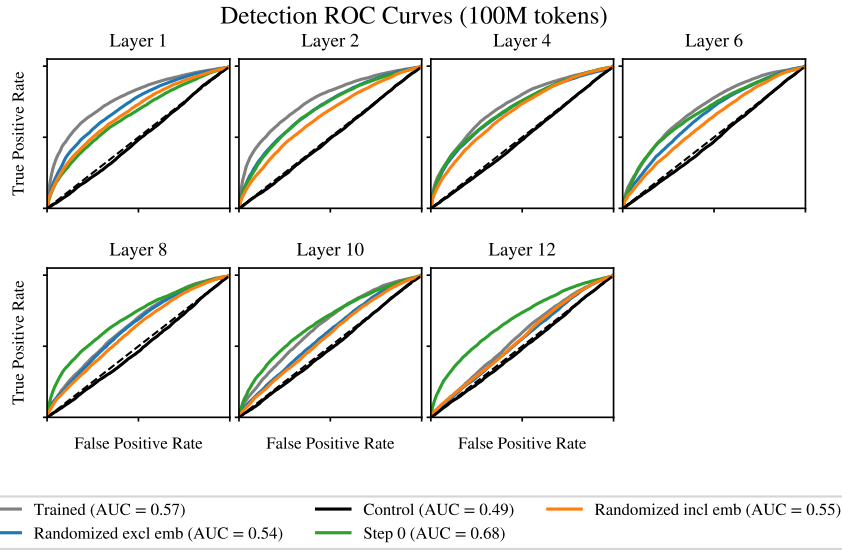


Figure 9: ‘Detection’ auto-interpretability ROC curves for Pythia-160m, computed over 100 SAE latents per layer. The SAE variants perform comparably, with a uniform degradation as the layer index increases, matching the ‘fuzzing’ result above.

B.3 PYTHIA 410M

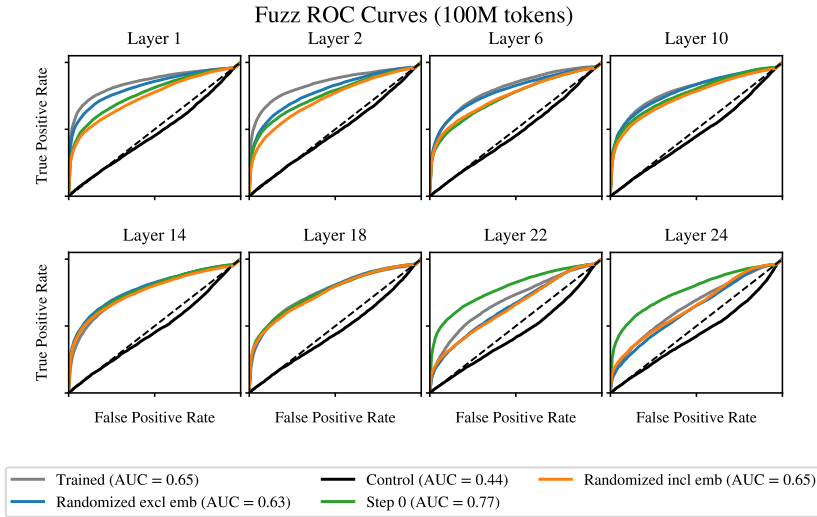


Figure 10: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-410m, computed over 100 SAE latents per layer. The variants perform comparably and degrade uniformly with layer index; here again the auto-interpretability scores do not distinguish the trained from the randomized models.

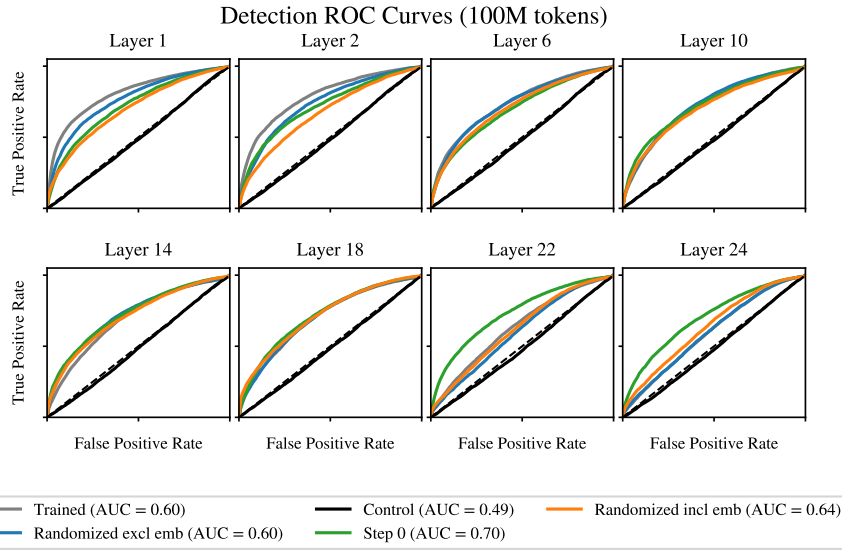


Figure 11: ‘Detection’ auto-interpretability ROC curves for Pythia-410m, computed over 100 SAE latents per layer. The SAE variants perform comparably, with a uniform degradation as the layer index increases, corroborating the ‘fuzzing’ result.

B.4 PYTHIA-1B

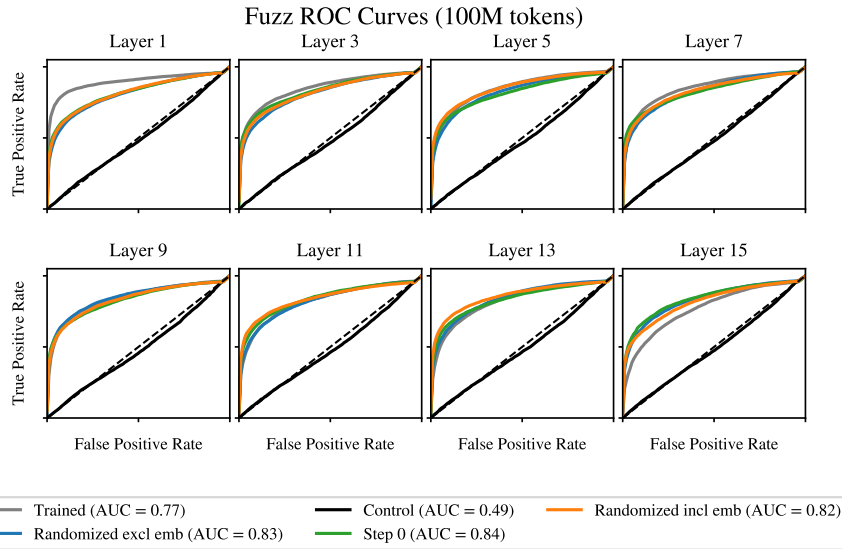


Figure 12: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-1b, computed over 100 SAE latents per layer. The SAE variants perform comparably; notably, and unlike the smaller models, this model shows no overall degradation with layer index—the onset of the scale-dependent behavior emphasized in the main text.

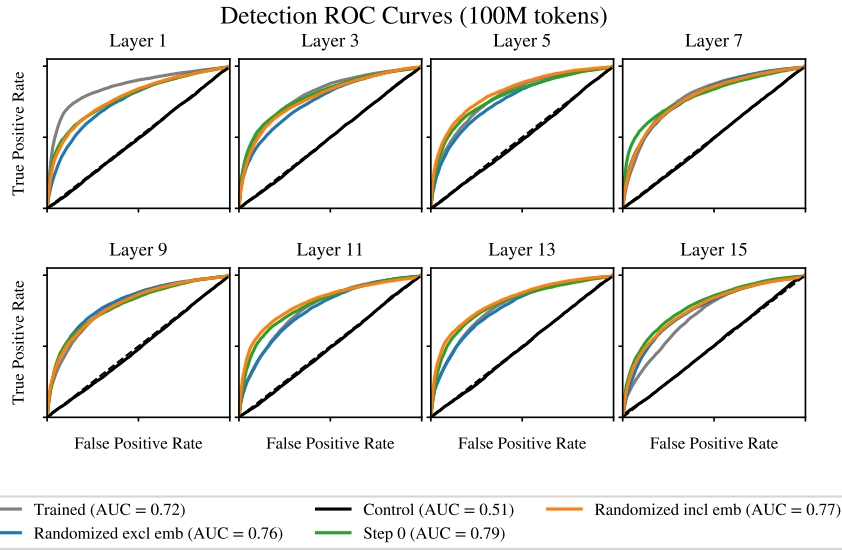


Figure 13: ‘Detection’ auto-interpretability ROC curves for Pythia-1b, computed over 100 SAE latents per layer. The SAE variants perform comparably and, as under ‘fuzzing,’ show no overall degradation with layer index.

B.5 PYTHIA 6.9B

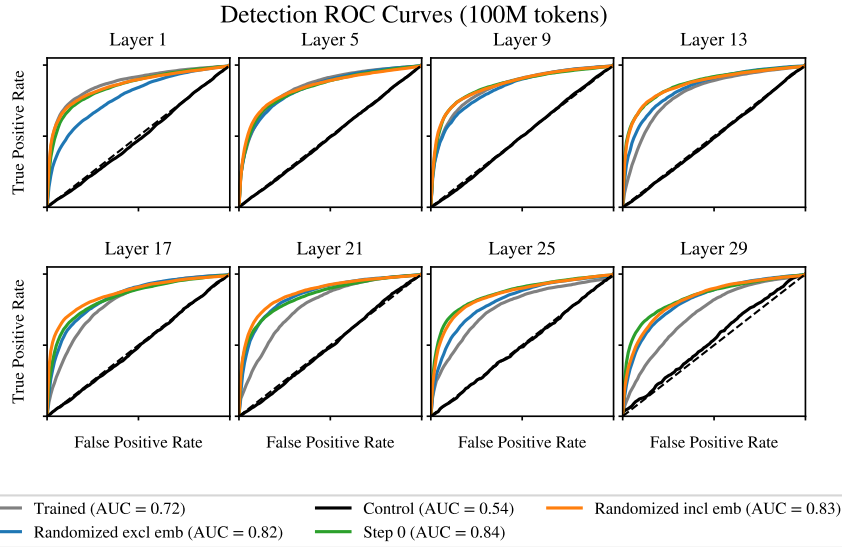


Figure 14: ‘Detection’ auto-interpretability ROC curves for Pythia-6.9b, computed over 100 SAE latents per layer. As with the ‘fuzzing’ curves of Figure 1, the SAE variants perform comparably to one another; the trained–randomized indistinguishability at the largest scale is thus reproduced under ‘detection’ scoring.

C EFFECT OF INCREASED TRAINING DATA

Our primary experiments trained SAEs on 100M tokens (Section 3). To rule out insufficient training data as an explanation for our findings, we repeated a subset of these experiments with SAEs trained on 1B tokens from the RedPajama dataset—a ten-fold increase. The results are materially unchanged (Figure 15), confirming that the trained-versus-random similarity is not an artifact of under-training the SAEs.

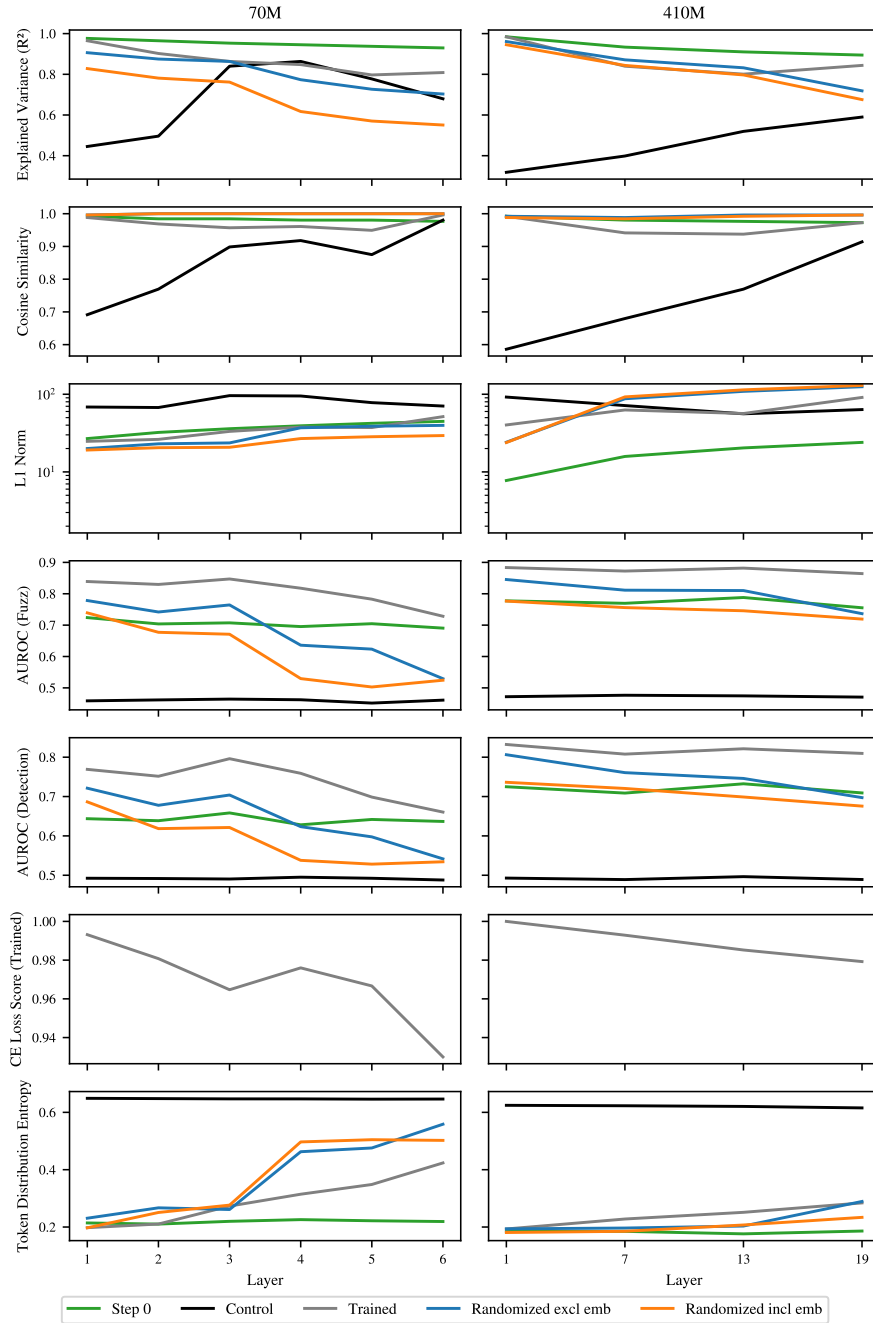


Figure 15: Evaluation metrics for SAEs trained on one billion tokens for the Pythia-70m and 410m models. These panels correspond directly to the matching columns of Figure 2, which reports the same metrics at 100M tokens; the behavior is qualitatively identical, establishing that the ten-fold increase in training data leaves the conclusions intact.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B

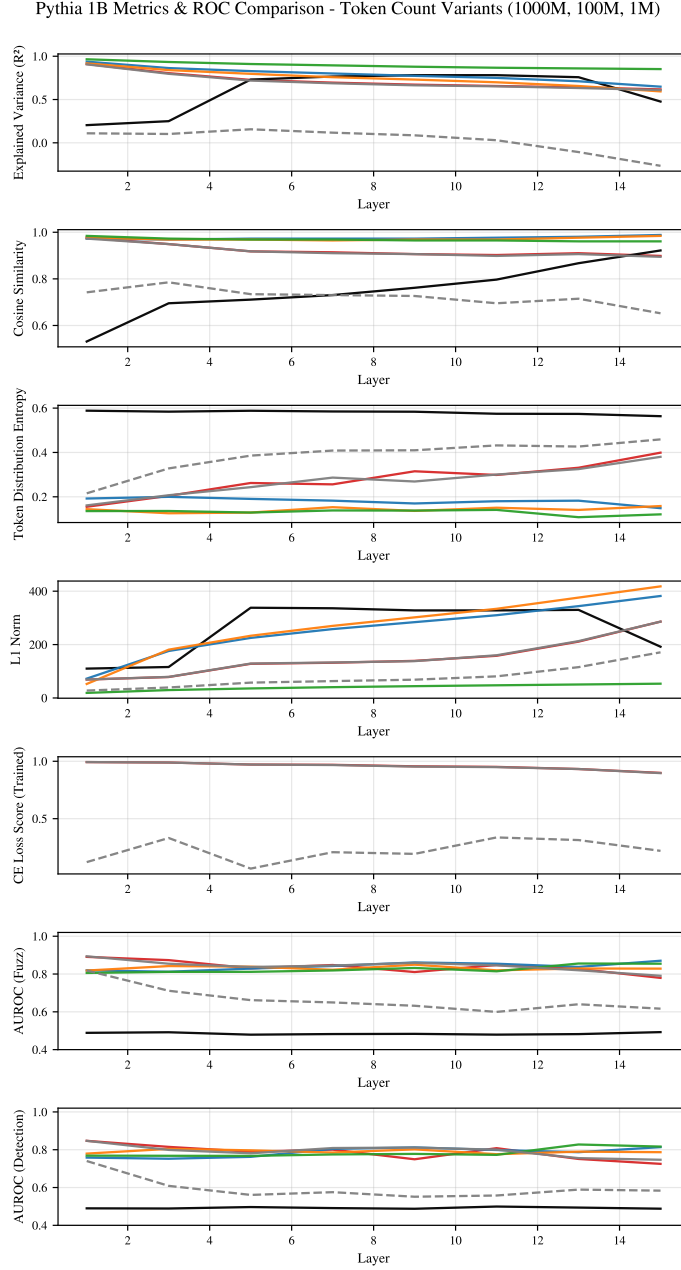
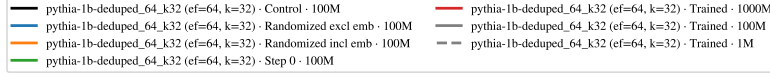


Figure 16: Evaluation metrics for SAEs trained on 1M and 1B tokens for Pythia-1b. At 1M tokens the explained variance and CE loss score are markedly lower, confirming that these SAEs are under-trained. Average auto-interpretability scores are only slightly lower at the earliest layers but decline sharply with depth. Even under this severe under-training, the layerwise trends in auto-interpretability and token-distribution entropy remain consistent with those of the other SAEs, indicating that the qualitative structure we report is robust to the training-data budget.

1242 E UNCERTAINTY PLOTS FOR PYTHIA-70M

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To characterize the variability of our evaluation metrics and confirm that the reported orderings are not seed-dependent, we recomputed them on Pythia-70m across five random seeds.

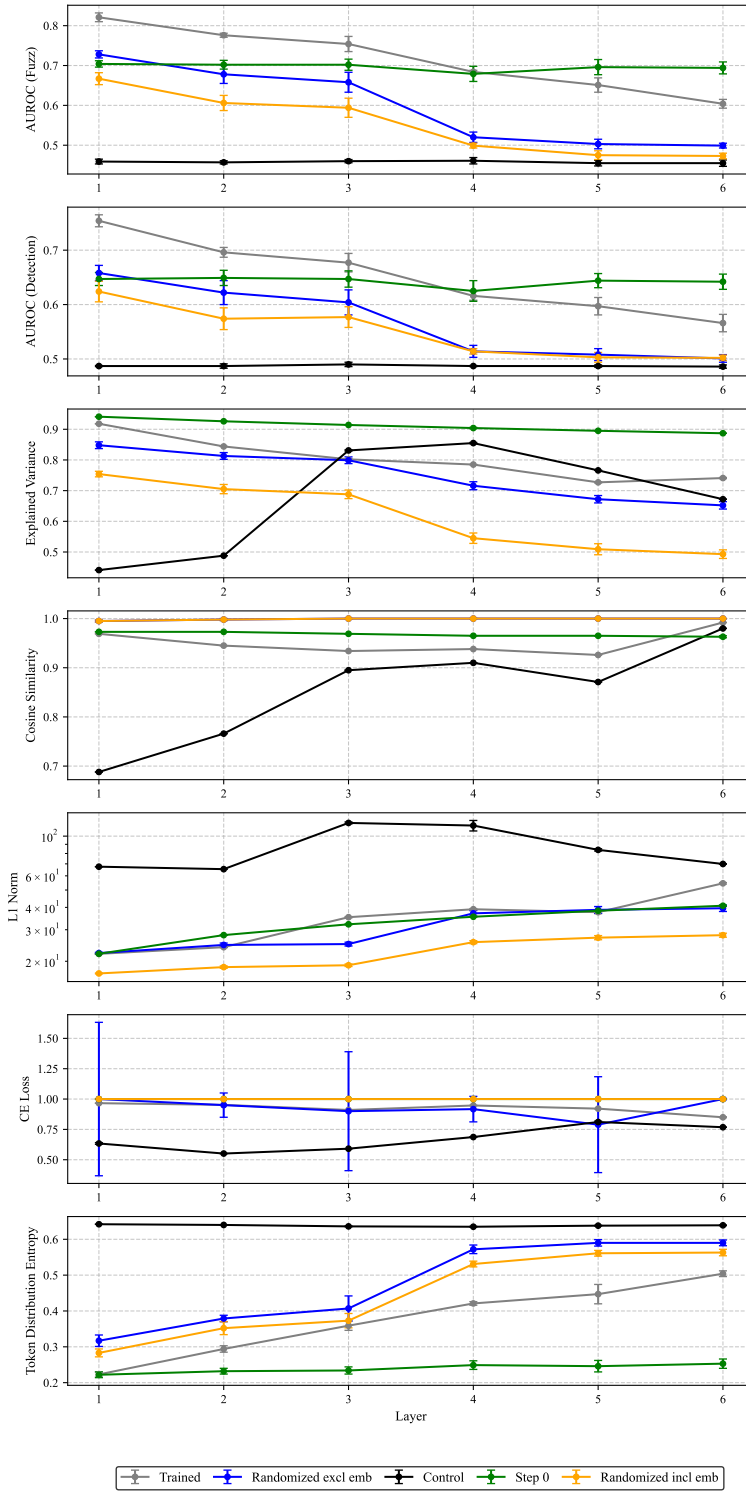


Figure 17: Uncertainty in the Pythia-70m evaluation metrics, computed across five random seeds. The variant orderings established in the main text persist once seed-to-seed variability is accounted for; the reported effects therefore exceed the sampling noise.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

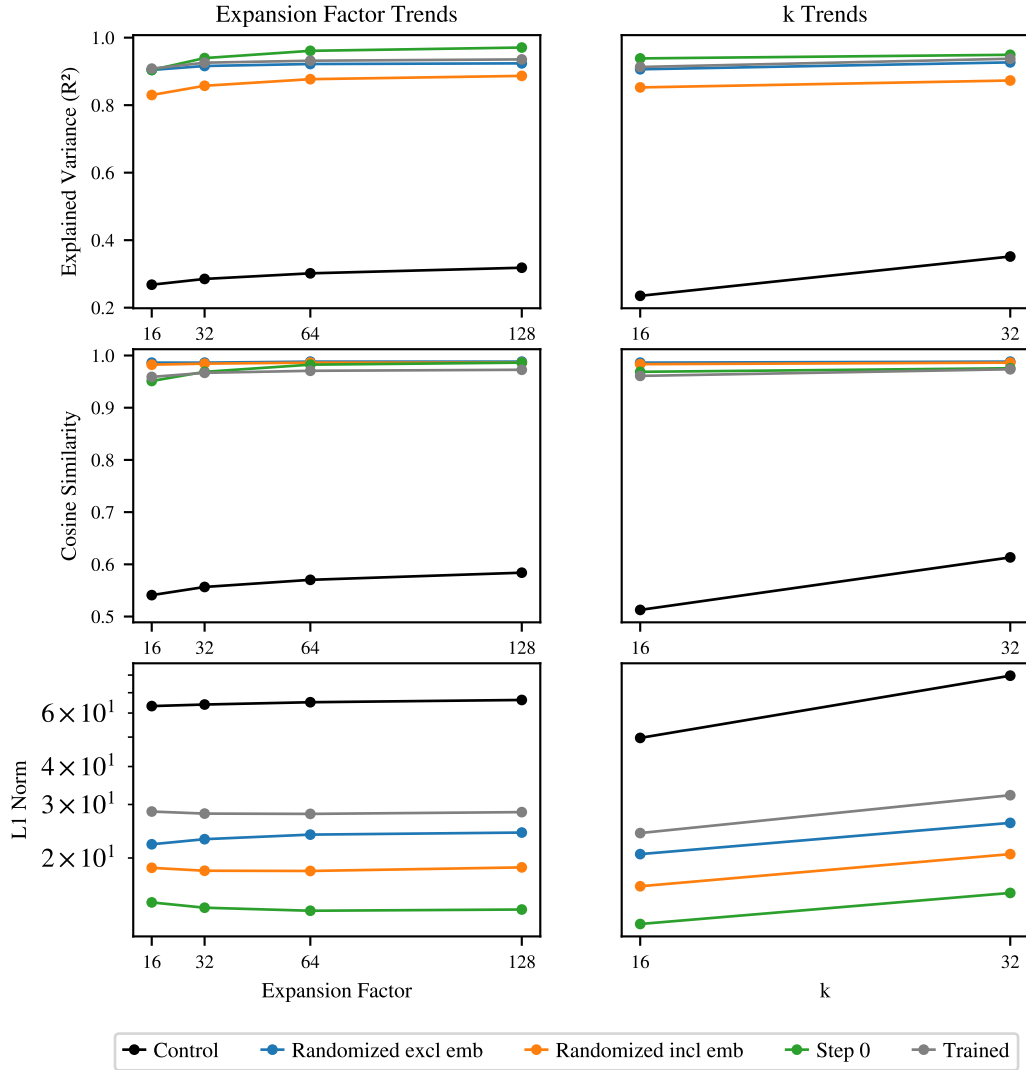


Figure 18: Robustness of SAE performance to hyperparameter selection on Pythia-160m. The standard evaluation metrics remain stable across expansion factors R from 16 to 128 and sparsities k of 16 and 32, and every initialization strategy retains its relative performance ordering throughout the sweep. This stability establishes both that moderate hyperparameter values (e.g., $R = 64$, $k = 32$) are sufficient and that our conclusions are not artifacts of a particular configuration.

G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B

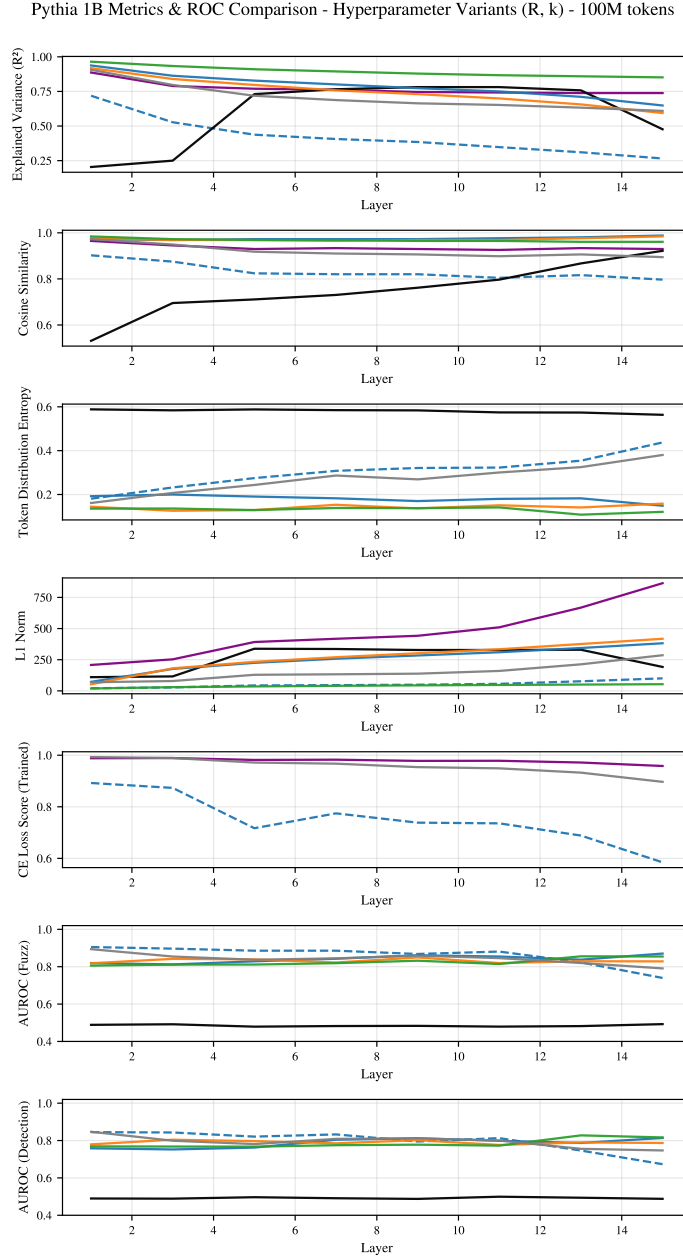


Figure 19: Evaluation metrics for SAEs trained on Pythia-1b under varying hyperparameters, alongside the main results from Figure 2. SAEs with a very small expansion factor $R = 2$ and sparsity $k = 4$ are clearly separated from our default configuration by explained variance and CE loss score. Even under this deliberately degraded configuration, their auto-interpretability scores remain close to those of the default-hyperparameter SAEs on both trained and randomized models—further evidence that the trained-versus-randomized similarity is a property of the auto-interpretability metric itself rather than an artifact of the hyperparameter choice.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

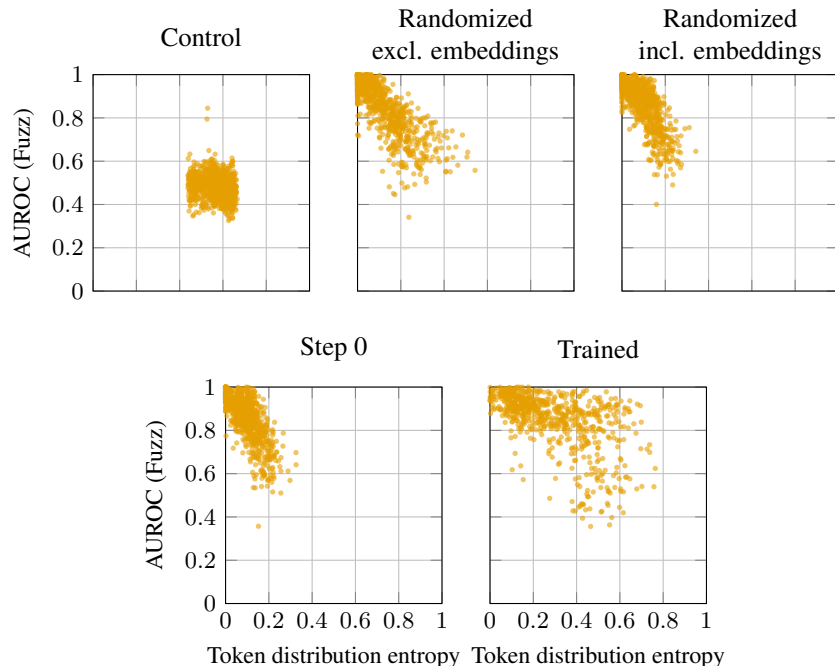


Figure 20: Per-latent token-distribution entropy against ‘fuzzing’ AUROC (auto-interpretability score) for SAEs trained on multiple layers of Pythia-6.9b. Each point is a single latent drawn from the same sample used to compute the aggregate metrics of Figures 1 and 2. Disaggregating to the latent level exposes structure that the aggregate scores conceal: the trained and randomized variants occupy visibly different regions of the entropy–AUROC plane even where their aggregate scores coincide.

Figure 20 separates the negative control, the randomized variants, and the trained variant cleanly:

- **Control:** latents exhibit uniformly high entropy—maximally activating examples whose activation patterns are spread across many tokens—together with low auto-interpretability scores, indicating generated explanations that fail to account for those patterns. The two quantities show no discernible correlation.
- **Randomized:** for each randomization scheme of Section 3, entropy and auto-interpretability are negatively correlated: the broader the range of tokens on which a latent fires, the less adequately its generated explanation accounts for its activation.
- **Trained:** the correlation is weaker, and, decisively, the trained variant alone populates a region that the randomized variants leave empty—latents with *both* high entropy and high auto-interpretability, whose activation patterns spread across multiple tokens yet remain consistent with the generated explanation.

This latent-level structure substantiates the central claim that aggregate auto-interpretability scores mask the difference between SAEs built on trained and randomized models. Randomized models with consistent token embeddings can produce easily explained ‘single-token’ features, but only transformers trained on natural language additionally yield the more complex, multi-token semantic features identified above—exactly the features that distinguish learned computation, and exactly the features that a token-distribution-entropy criterion isolates.

I A TOY MODEL OF SUPERPOSITION

In Section 4 we trained SAEs on toy data constructed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). This section specifies the data-generation procedure and the training setup in full, so that both are exactly reproducible.

I.1 DATA GENERATION

We first construct ground-truth features by sampling n_s points on an n_d -dimensional hypersphere.

The feature coefficients for each sample are then obtained through the following sequence of operations: generating $A \in \mathbb{R}^{n_s \times n_s}$ where $A_{ij} \sim \mathcal{N}(0, 1)$, defining a covariance matrix $\Sigma = AA^\top$, sampling $\vec{\alpha} \in \mathbb{R}^{n_s}$ where $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$, projecting α_i onto the c.d.f. of $\mathcal{N}(0, 1)$, decaying $\alpha_i \rightarrow \alpha_i^{\lambda_i}$ where $\lambda \in \mathbb{R}$, normalizing $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ where $m \in \mathbb{R}$, and performing n_s independent Bernoulli trials with $p = \alpha_i$. The trial outcomes are finally multiplied by n_s independent samples from a continuous uniform distribution $\mathcal{U}_{[0,1]}$, yielding the feature coefficients for the sample.

The two free parameters of this construction admit direct interpretations: λ governs how sharply the frequency of nonzero ground-truth feature coefficients decays with the feature index i , while m sets the expected number of nonzero feature coefficients per sample and thus the overall sparsity level.

Following Sharkey et al. (2022), we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$. A Python implementation of the full procedure is given in Figure 21.

I.2 TRAINING

The SAEs of Section 4 consist of a linear encoder with a bias term, a ReLU activation, and a linear decoder without a bias term. The decoder weights are initialized orthogonally, and the decoder weight vectors are renormalized to unit norm before every training step.

The training loss is the mean squared error (MSE) between the input and decoded vectors, augmented by the mean L^1 norm of the encoded vectors scaled by a penalty coefficient that we sweep between 1×10^{-3} and 100 to trace the Pareto frontiers reported in the main text.

We train for 100 epochs in every configuration: on 10K data points across 10 random seeds for the toy data, and on 400K data points with a single random seed for the word vectors. In both cases we hold out 10% of the data points as a validation set, and all reported evaluation metrics are computed on this held-out set.

The MLPs of Section 4 comprise two layers (a single hidden layer) with a ReLU activation. Their input and output dimensions are both n_d and their hidden dimension is $4n_d$ —choices loosely modeled on the feed-forward components of transformer language models.

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1578 1 def generate_sharkey(
1579 2     num_samples: int,
1580 3     num_inputs: int,
1581 4     num_features: int,
1582 5     avg_active_features: float,
1583 6     lambda_decay: float,
1584 7 ) -> tuple[Tensor, Tensor]:
1585 8     """
1586 9     Args:
1587 10         num_samples (int): The number of samples to generate.
1588 11         num_inputs (int): The number of input dimensions.
1589 12         num_features (int): The number of ground truth features.
1590 13         avg_active_features (float): The average number of
1591 14             ground truth features active at a time.
1592 15         lambda_decay (float): The exponential decay factor for
1593 16             feature probabilities.
1594 17     """
1595 18     features = torch.randn(num_inputs, num_features)
1596 19     features /= torch.norm(features, dim=0, keepdim=True)
1597 20
1598 21     covariance = torch.randn(num_features, num_features)
1599 22     covariance = covariance @ covariance.T
1600 23     correlated_normal = MultivariateNormal(
1601 24         torch.zeros(num_features), covariance_matrix=covariance
1602 25     )
1603 26
1604 27     samples = []
1605 28     for _ in range(num_samples):
1606 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1607 30         p = p ** (lambda_decay * torch.arange(num_features))
1608 31         p = p * (avg_active_features / (num_features * p.mean()))
1609 32         p = torch.bernoulli(p.clamp(0, 1))
1610 33         coef = p * torch.rand(num_features)
1611 34
1612 35         sample = coef @ features.T
1613 36         samples.append(sample)
1614 37
1615 38     return torch.stack(samples), features

```

Figure 21: Python implementation of the data-generation procedure introduced by Sharkey et al. (2022) and used throughout Section 4, provided to make the construction exactly reproducible.

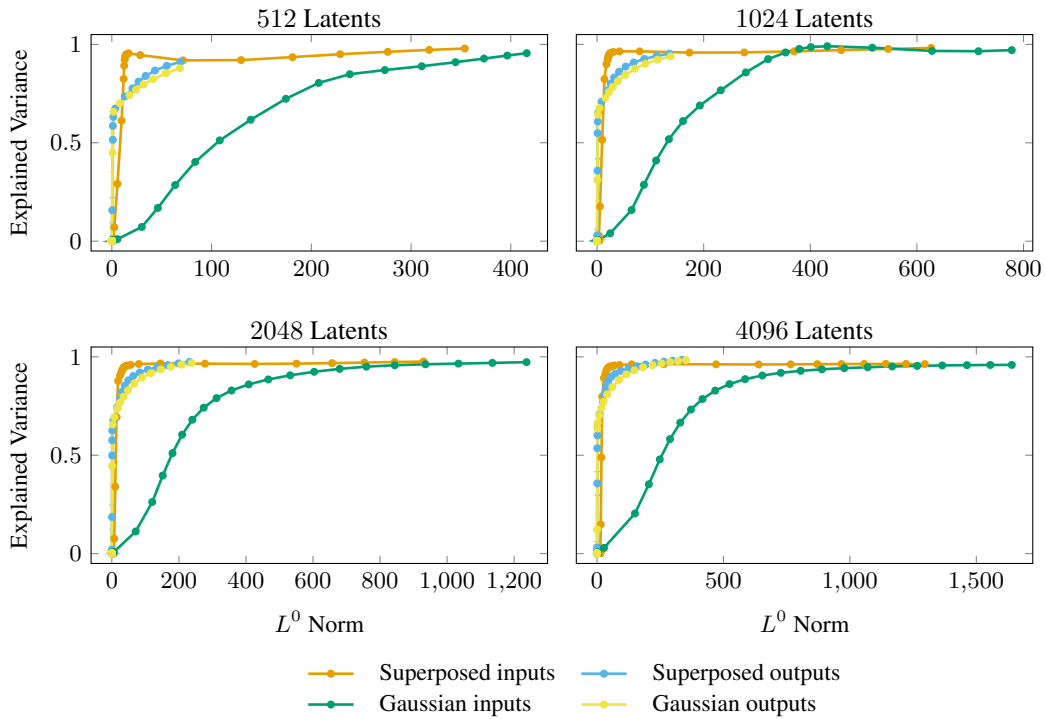


Figure 22: Pareto frontiers of explained variance against sparsity, here measured by the L^0 norm, for toy datasets generated to exhibit superposition, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The random MLP shifts the frontier toward greater sparsity under this measure as well.

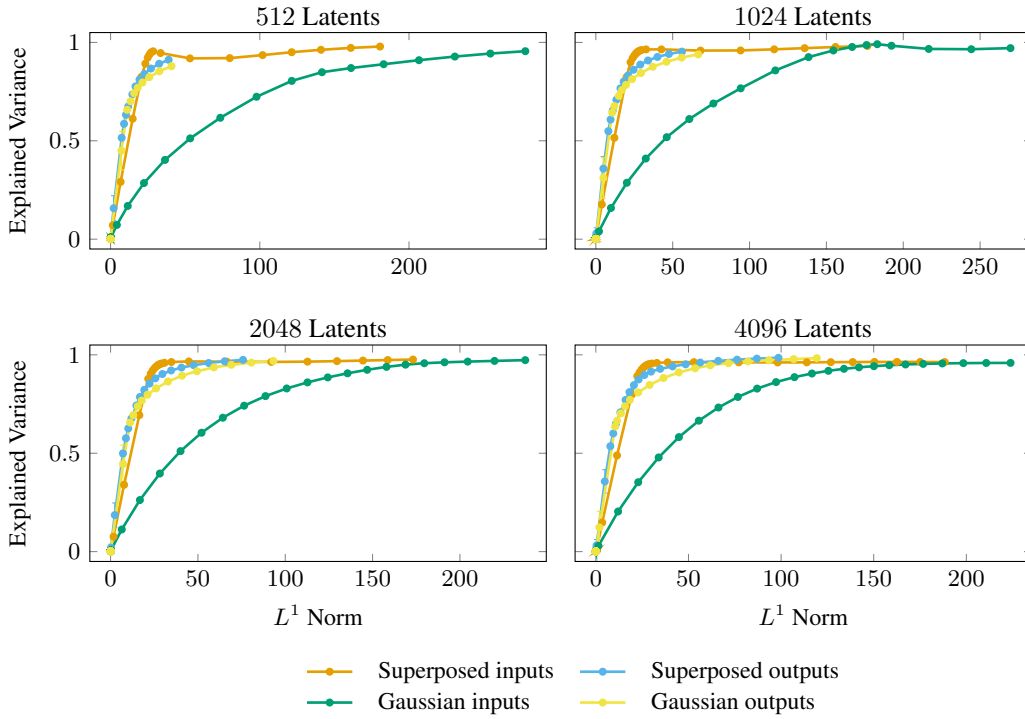


Figure 23: Pareto frontiers of explained variance against sparsity, here measured by the L^1 norm, for toy datasets generated to exhibit superposition, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The sparsification effect is preserved under the L^1 measure.

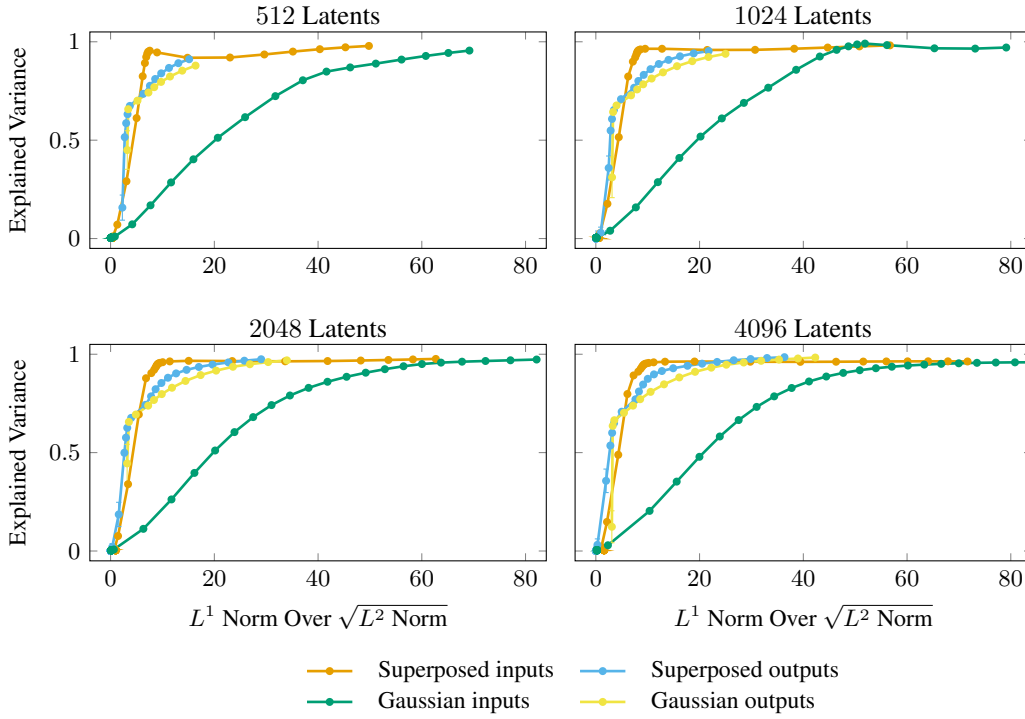


Figure 24: Pareto frontiers of explained variance against sparsity, here measured by the L^1 norm divided by the square root of the L^2 norm, for toy datasets generated to exhibit superposition, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. This is the scale-invariant sparsity measure reported in the main text (Figure 5a).

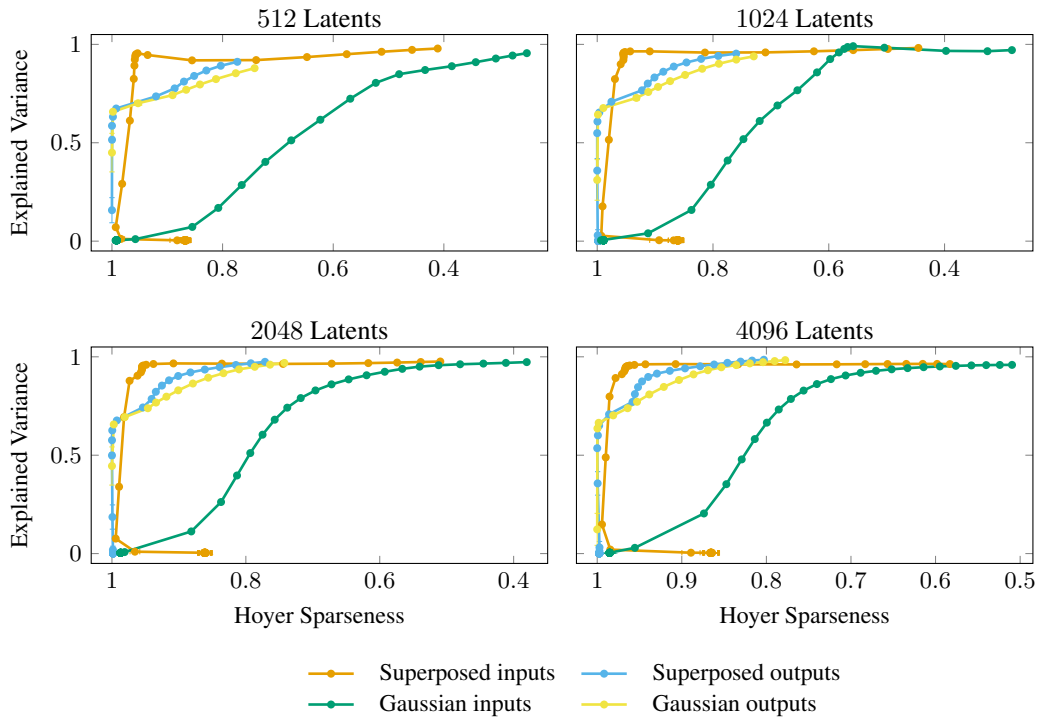


Figure 25: Pareto frontiers of explained variance against sparsity, here measured by the Hoyer sparseness, for toy datasets generated to exhibit superposition, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The same ordering of the frontiers holds under this fourth measure, so the conclusion does not depend on the particular sparsity metric.

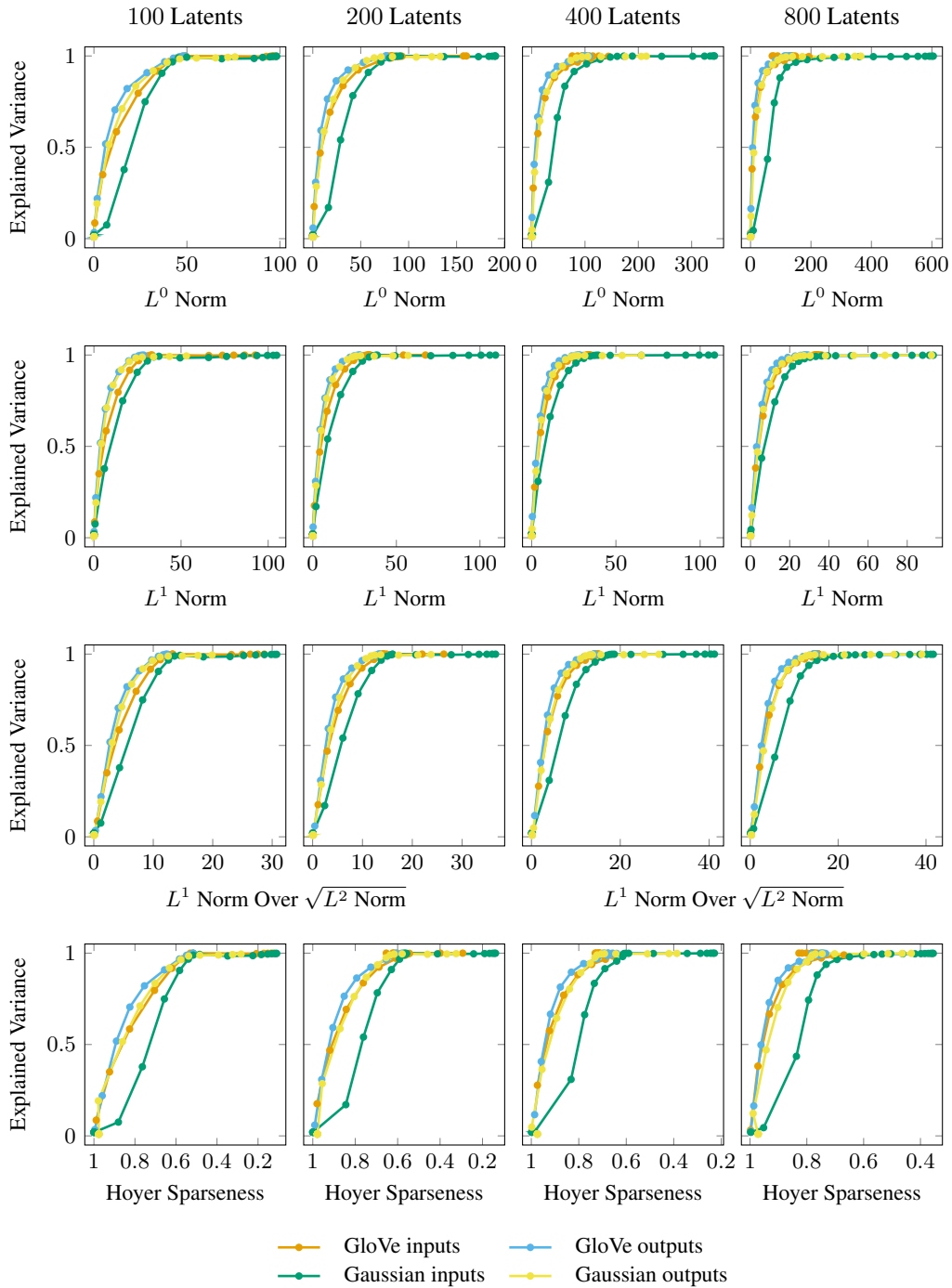


Figure 26: Pareto frontiers of explained variance against several sparsity measures for 50-dimensional GloVe word vectors, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. Passing the embeddings through the random MLP again increases their apparent sparsity.

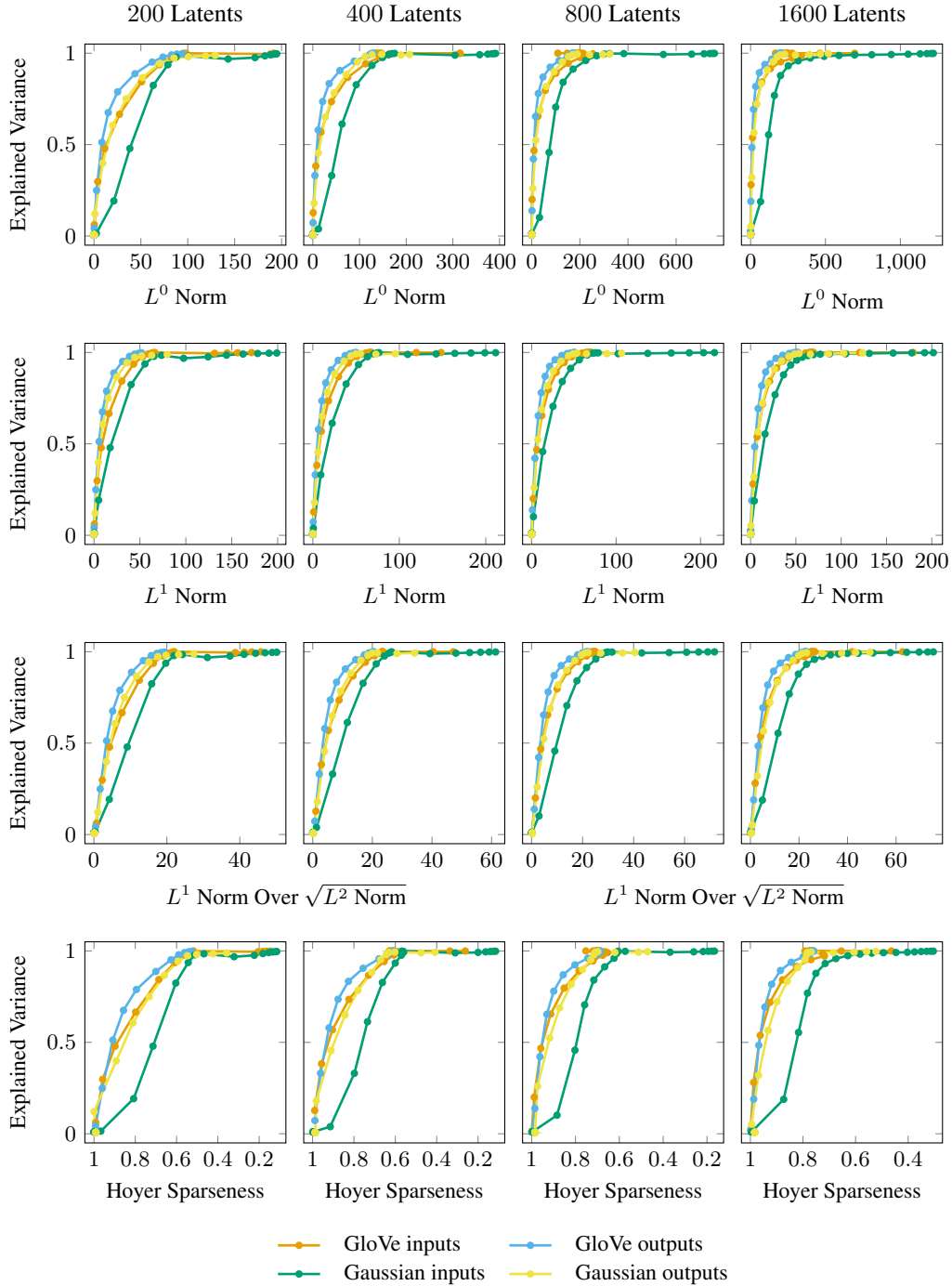


Figure 27: Pareto frontiers of explained variance against several sparsity measures for 100-dimensional GloVe word embeddings, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The effect observed at 50 dimensions persists as the embedding dimension increases.

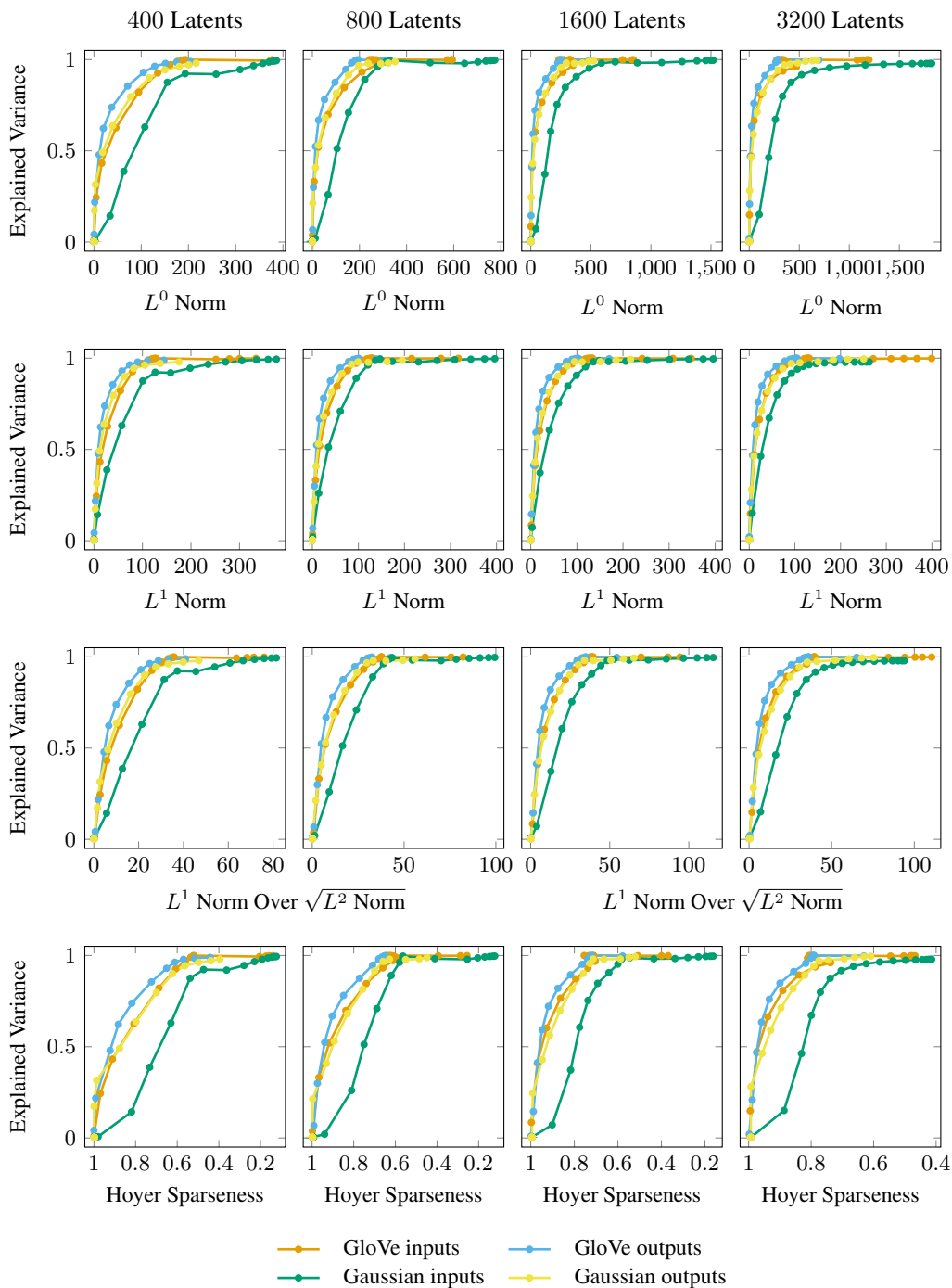


Figure 28: Pareto frontiers of explained variance against several sparsity measures for 200-dimensional GloVe word embeddings, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The sparsification effect remains evident at 200 dimensions.

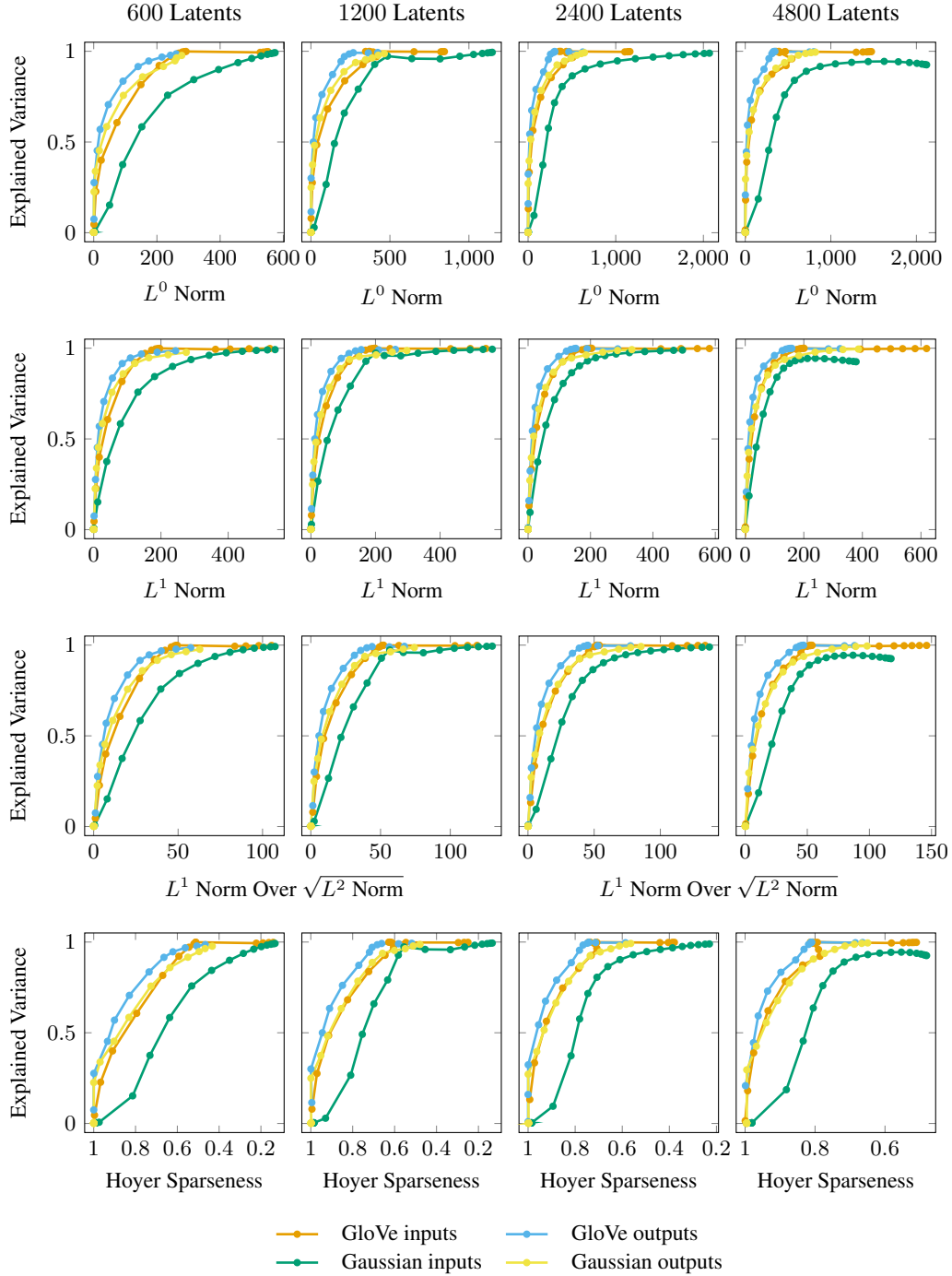


Figure 29: Pareto frontiers of explained variance against several sparsity measures for 300-dimensional GloVe word embeddings, for Gaussian controls matched in mean and variance, and for the corresponding outputs when each is passed through a randomly initialized two-layer MLP. The 300-dimensional case is the one summarized in the main text (Figure 5b); the effect holds across every embedding dimension we examine.

J EXAMPLE FEATURES FOR PYTHIA-6.9B

VARIANT: TRAINED

FEATURE 180935 (LAYER 0)

Interpretation: The term "security" is predominantly used to refer to protection, safety, and measures to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or quantification, although "oz" appears less frequently and often in a different context.

Top Examples:

1. Text: <endoftext>—ζ—. If for any reason you are unhappy with our service please contact us directly so we can make it right for you. Journal of Cyber Security, Vol.
Activation: 4.3750
Active tokens: Security
2. Text: <endoftext>—ζ— Security Practitioner. Tremendously passing CompTIA Advanced Security Practitioner (casp) cert has never been as easy as it
Activation: 4.3125
Active tokens: Security Security
3. Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to find out more. Technically proficient using
Activation: 4.3125
Active tokens: Security

FEATURE 93790 (LAYER 8)

Interpretation: Nouns and phrases related to economic concepts, development, and business, often referring to growth, progress, and improvement.

Top Examples:

1. Text: training requirements. See "Workforce" section for additional information. The Economic Development Transportation Fund, commonly referred to as the "Road Fund," is an
Activation: 21.3750
Active tokens: Development
2. Text: Montréal. The Williamsburg Economic Development Authority offers a 33% matching grant up to \$7,500 for exterior improvements to existing businesses in the City of
Activation: 20.2500
Active tokens: Development Authority
3. Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic Development Corporation was "circumventing" laws with its restructuring. In
Activation: 20.0000
Active tokens: Development

FEATURE 128309 (LAYER 12)

Interpretation: Various types of punctuation and grammatical elements that separate words or phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connections, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses or ideas.

Top Examples:

1. Text: Run it in JDK6, and it will print "[axons, bandrils, chumblies]". If you are having trouble switching from
Activation: 8.0000
Active tokens: in JD K

2106 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
 2107 of 3-dimensional spaces lead us to the setting for our study
 2108 2. Activation: 7.8125
 2109 Active tokens: □
 2110 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
 2111 If you are looking for a simple and
 2112 3. Activation: 7.7188
 2113 Active tokens: . path expand user

2114
 2115
 2116
 2117 VARIANT: STEP 0

2118
 2119 FEATURE 126848 (LAYER 12)

2120
 2121 **Interpretation:** Nouns denoting people who train others, units or marks of measurement, and
 2122 abbreviations or acronyms representing specific standards or technologies.

2123 **Top Examples:**

- 2124
 2125 Text: What are the various lessons a member can access at a tennis club? Whether you are a
 2126 1. beginner or advanced player, trainers help you to choose the right gaming
 2127 Activation: 13.1250
 2128 Active tokens: trainers
 2129 Text: report include various simulation platforms and Serious Games. The report also analyzes
 2130 some major allied products such as patient simulators and task trainers. The technologies
 2131 2. analyzed
 2132 Activation: 13.0625
 2133 Active tokens: trainers
 2134 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
 2135 Asics trainers. We've got numerous styles from
 2136 3. Activation: 13.0000
 2137 Active tokens: trainers
-

2138
 2139
 2140 FEATURE 2125 (LAYER 4)

2141
 2142 **Interpretation:** The word "papers" is often used in contexts referring to written documents, such as
 2143 academic papers, court documents, or printed materials, and is frequently mentioned in relation to
 2144 tasks like writing, research, and education.

2145 **Top Examples:**

- 2146
 2147 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
 2148 1. Sierra leone. Taiwan. Turkey. Venezuela.
 2149 Activation: 5.7188
 2150 Active tokens: papers
 2151 Text: that can be associated with interaction with other individuals. For everybody who is
 2152 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
 2153 2. proofreading services
 2154 Activation: 5.6875
 2155 Active tokens: papers
 2156 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
 2157 RC, licence and insurance coverage as effectively as PUC (
 2158 3. Activation: 5.6562
 2159 Active tokens: papers
-

2160 FEATURE 9944 (LAYER 16)

2161

2162 **Interpretation:** Words or parts of words that are usually the beginning or end of a proper noun,

2163 surname, or a word of foreign origin.

2164 **Top Examples:**

2165

2166 Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go

2167 1. Now? Where Do Music Festivals Go Now? Are you ready

2168 Activation: 10.4375

2169 Active tokens: Fest Fest

2170 Text: client streams, and encouraging existing customers to become more involved.Welcome to

2171 2. Fil Fest USA!! Do you love lumpia? Can you eat a handful of them

2172 Activation: 10.3750

2173 Active tokens: Fest

2174 Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!:

2175 3. Off to Comic Fest 2012. Should be interesting if nothing else!

2176 Activation: 10.3750

2177 Active tokens: Fest Fest

2178

2179

2180 VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS

2181

2182 FEATURE 151030 (LAYER 28)

2183 **Interpretation:** Common nouns, proper nouns, or adjectives found in various contexts, including but

2184 not limited to geographical locations, people, organizations, time, and concepts, often possessing

2185 relevance to the surrounding text.

2186 **Top Examples:**

2187

2188 Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994,

2189 1. <http://articles.latimes.com/1994-06-12>

2190 Activation: 58.0000

2191 Active tokens: lat

2192 Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique

2193 2. a distinct blunt in a retailer you superlativeness be

2194 Activation: 56.5000

2195 Active tokens: lat

2196 Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over

2197 3. six days in 1971, as he struggles

2198 Activation: 55.7500

2199 Active tokens: lat lat

2200

2201

2202 FEATURE 98924 (LAYER 12)

2203 **Interpretation:** Adjectives describing size, or nouns representing concepts or objects that are being

2204 described in terms of their size.

2205 **Top Examples:**

2206

2207 Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to

2208 1. the left is for small dogs only. Troup 69

2209 Activation: 46.0000

2210 Active tokens: small

2211 Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication????

2212 2. Or I am really need a cool,small

2213 Activation: 45.5000

Active tokens: small

2214 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
2215 3. Activation: 44.7500
2216 Active tokens: small

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FEATURE 180589 (LAYER 24)

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Interpretation: Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in a professional or organizational context, sometimes accompanied by proper nouns and a few instances with words having suffixes or prefixes.

Top Examples:

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VARIANT: RANDOMIZED INCLUDING EMBEDDINGS

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2246

2247

FEATURE 39748 (LAYER 0)

Interpretation: Words contain “Pul” are often used in the context of Pulitzer, a prestigious journalism award, while “Looking” typically precedes a phrase expressing anticipation, expectation, or searching for something.

Top Examples:

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FEATURE 15633 (LAYER 20)

2265

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2267

Interpretation: Nouns representing individuals or entities possessing or having authority over something, often in a possessive or authoritative relationship with that thing.

Top Examples:

-
- 2268 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
 2269 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
 2270 1. Activation: 80.0000
 2271 Active tokens: Stark
- 2272 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
 2273 system used, ETC's Paradigm architectural control, is
 2274 2. Activation: 77.5000
 2275 Active tokens: Stark
- 2276 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
 2277 throughout May, just check with your local shop to guarantee your place
 2278 3. Activation: 77.5000
 2279 Active tokens: Stark
-

2280
 2281
 2282 FEATURE 6069 (LAYER 4)

2283
 2284 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
 2285 connotations often related to religion or technology.

2286 **Top Examples:**

- 2287
 2288 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
 2289 and a collection of Hans Christian Anderson fairy
 2290 1. Activation: 22.1250
 2291 Active tokens: Christian
- 2292 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
 2293 So yeah, Christian leaders won't make any big
 2294 2. Activation: 22.0000
 2295 Active tokens: Christian
- 2296 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
 2297 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into
 2298 3. Activation: 22.0000
 2299 Active tokens: Christian
-

2300
 2301
 2302 VARIANT: CONTROL

2303
 2304 FEATURE 290 (LAYER 4)

2305
 2306 **Interpretation:** Function words and occasionally nouns or proper nouns that seem to be emphasized
 2307 as part of a larger phrase or topic, often indicating transition or conjunction.

2308 **Top Examples:**

- 2309 Text: <endoftext—¿—ance - Chapters: 1 - Words:. Fruits Basket - Rated: T - English -
 2310 Romance/Angst - Chapters:
 2311 1. Activation: 5.0938
 2312 Active tokens: -
- 2313 Text: ococcus neoformans-reactive and total immunoglobulin profiles of human immunodeficiency virus-infected and uninfected Ugandans'. Clinical and Diagnostic Laboratory Immunology, Vol 12
 2314 2. Activation: 5.0938
 2315 Active tokens: un
- 2316 Text: and in fact any correspondence that the social club had in the run-up to the sit-in was from
 2317 the social club's own solicitors.
 2318 3. Activation: 5.0625
 2319 Active tokens: the
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 2321
-

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FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

All experiments were run on a single NVIDIA A100 80GB GPU in a private cluster. Table 1 reports the approximate wall-clock duration of the final experiments, broken down by model size and transformer variant. We estimate the combined cost of preliminary and failed experiments to be roughly equal to that of the final experiments, so the total compute expended is approximately twice the figure reported below.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate compute time required for our experiments, by model size and variant, totaling roughly 435 GPU-hours on a single A100 80GB.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

We provide more detailed ‘feature dashboards’ that report, for each feature, its activation patterns, its token-distribution entropy, and its auto-interpretability (‘fuzz’ ROC) score together in one view. For each of the control, randomized, and trained variants of Section 3, trained on every fourth layer of Pythia-6.9b, we show two randomly sampled features, so that the qualitative differences between variants can be inspected directly rather than only in aggregate.

L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

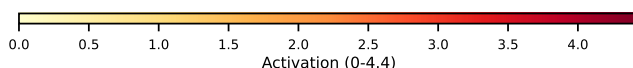
Ex 1 which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with

Ex 2 Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the

Ex 3 the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,

Ex 4 treated equal. What was the Statue of Liberty originally used for? Shadows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition

Ex 5 by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave



Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

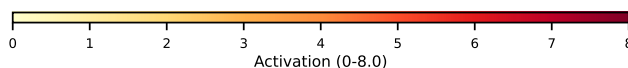
Ex 1 <|endoftext|>**intin** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar

Ex 2 the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.

Ex 3 near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked

Ex 4 least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**

Ex 5 than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked

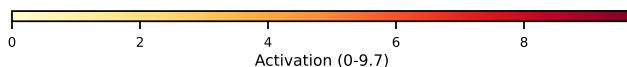


Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

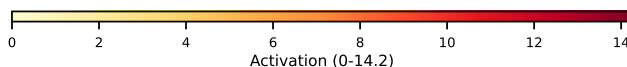


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

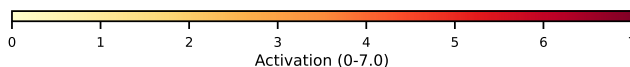
Ex 1 . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■

Ex 2 bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from

Ex 3 the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer

Ex 4 breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor

Ex 5 whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly



Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

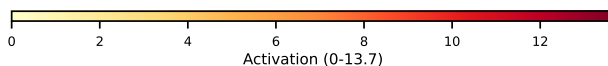
Ex 1 fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you ■are in trusted hands in Bussey

Ex 2 In addition, ■you?ll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure ■have an unforget

Ex 3 of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you

Ex 4 . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.

Ex 5 in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

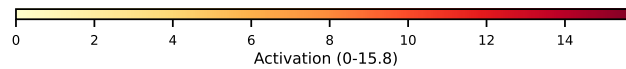
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `="scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

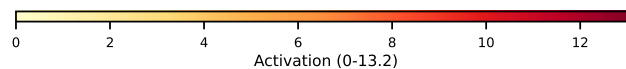
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidad de ho`

Ex 2 `<|endoftext|>bas separado del entorno de ... bueno pues quer237;a saber si el alcohol de limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le lien de confiance! Je viens de m'insc`

Ex 4 `<|endoftext|>water relationship. Rsum. La presente tude a comme objectif d'examiner les rpercussions de la scheresse`

Ex 5 `<|endoftext|>más nada. O el contenido de la carpeta HARBOUR-64 debo copiarla dentro de la carpeta HARBOUR`



Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

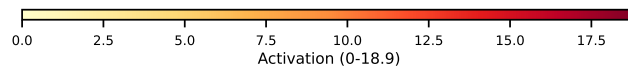
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risskov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

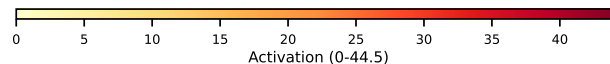
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A.J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

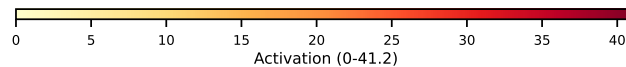


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

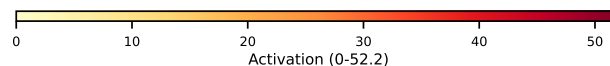


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <|endoftext|> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

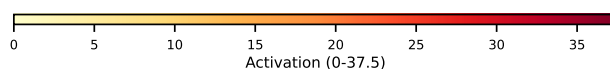


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served **include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (**
- Ex 2** Rs 136 crore having facilities for processing **milk** and **milk products** besides **packaging vegetables and fruits**. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** **ori Academy** and **Daycare**, **Little Kickers**, **Fourth Trafalgar Scouts**, **community yoga** and **volleyball**. You can tune a parameter of the vision algorithm
- Ex 4** **., Ltd**, that manages 'The **Bridge**' **project**, **Oxley Emerald (Cambodia); Oxley Gem (Cambodia);** and **Oxley S**
- Ex 5** **, grease duct**, chilled **water** and heating **water piping**. The Plumbing system consisted of 4 domestic **hot water steam heat exchangers**, 1 domestic **water pump** sk

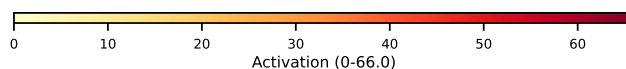


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, **note that I am** building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. **(Note: we**
- Ex 3** your own (depending on what interests you most). Also, **note that it is** a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. **PS: Note that** this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). **Note that** the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

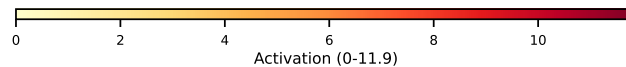
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

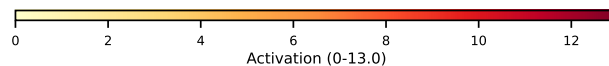
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

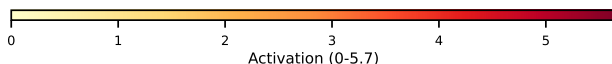
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

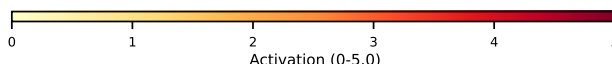
Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach



Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

Ex 1 support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque

Ex 2 pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of

Ex 3 for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of

Ex 4 their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,

Ex 5 Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention



Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

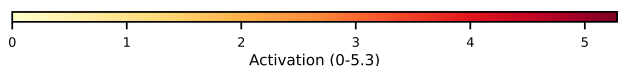
Ex 1 See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world

Ex 2 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts

Ex 3 an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick

Ex 4 <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur

Ex 5 .0 ml/min. at a total pressure of 0. **5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the



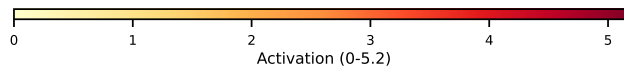
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2998
2999
3000
3001
3002
3003
3004
3005
3006
3007
3008
3009
3010
3011
3012
3013
3014
3015
3016
3017
3018
3019
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3021
3022
3023

Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

- Ex 1** y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds Tags bluebirds Comments on Bluebirds hanging out with Frosty!
- Ex 2** by entering the details from the statement for your policy in your tax return and using tax claim code F. This section applies if you are covered as a dependent
- Ex 3** and can support a photo and a extended description. Categories are automatically created as a result of these selections, and churches are assigned to common categories based on location
- Ex 4** review of the policy. Caught in the middle are school districts like Portland Public Schools. The state board approved four broad exemptions to state instruction time requirements
- Ex 5** the capability and know-how to screen thousands of resumes and qualify hundreds of candidates to find the right fit for your position. Through our detailed screening process,

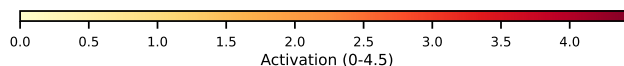


Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

- Ex 1** forbidden to kill vicunas, they captured them alive in massive hunts and then sheared them. Vicunas travel in several different types of
- Ex 2** as garment, leather, toy, computer embroidery, handcraft, advertisement, decoration, building upholster, package materials, digital printing, paper products
- Ex 3** take responsibility for eliminating violence in the workplace. Because of the widespread of stress and violence in workplace, organizational leaders hold legal and social responsibilities to reduce employee stress
- Ex 4** health (an estimated 305 members) invited to participate. Response frequencies were analysed in SPSS. Open ended comments were subjected to thematic analysis. Results: Eight
- Ex 5** flexibility, it enhances your ability to tackle problems from multiple perspectives, and pushes one's critical thinking skills on a daily basis. Growing up in an international



Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

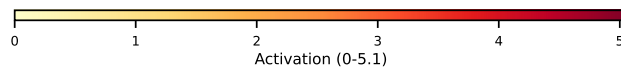
Ex 1 we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!

Ex 2 its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as

Ex 3 uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your

Ex 4 Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest

Ex 5 <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common



Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

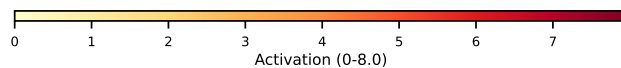
Ex 1 do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **■** neutral knits

Ex 2 direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments

Ex 3 <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that

Ex 4 more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine.Good free wifi internet access throughout hotel Very close to

Ex 5 **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

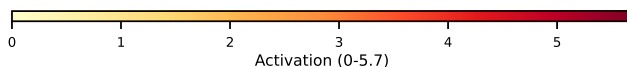
Ex 1 manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru

Ex 2 then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and

Ex 3 between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions

Ex 4 great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and

Ex 5 that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their



Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

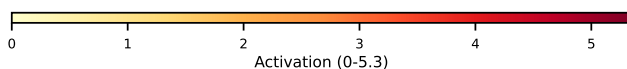
Ex 1 <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda

Ex 2 **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned

Ex 3 groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great

Ex 4 phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can

Ex 5 ope – its new editorial brand focused on health. Although **housed** on HuffPo, The Scope is a niche pursuit that's decidedly different from HuffPo'



Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

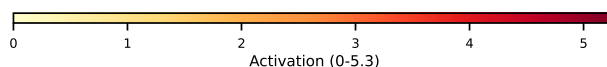
Ex 1 <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like

Ex 2 <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War

Ex 3 <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to

Ex 4 <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways

Ex 5 <|endoftext|> ! The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un



Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

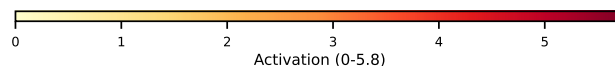
Ex 1 United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of

Ex 2 with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"

Ex 3 cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20

Ex 4 Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.

Ex 5 in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

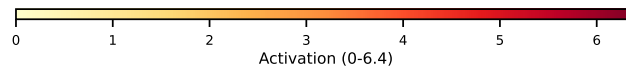
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

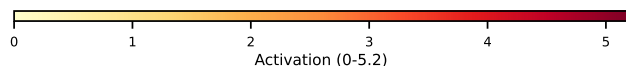
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 t' c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat this **MEAT**,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educational efforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in



Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

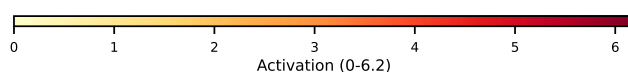
Ex 1 <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in **a** stage play co-produced by Bunuba Films and the Black Swan

Ex 2 <|endoftext|> in Human Resources Management: An Assessment of **Human** Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:

Ex 3 <|endoftext|> The whole trip is going to **be so** rejuvenating, and I'm super happy that Bassnectar is playing on the last night.

Ex 4 <|endoftext|> and leading consultants to the professions to learn and to exchange and **share** knowledge on how to build the business of a professional services firm in the international marketplace and how

Ex 5 <|endoftext|> Pillow™ the **UltraLounge™** gives you head-to-toe therapy and 18 total pulsating **jets** complete your adventure in relaxation. Length



Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

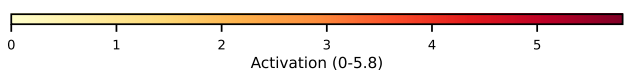
Ex 1 <|endoftext|> **al.** for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288

Ex 2 <|endoftext|> **it** comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this

Ex 3 <|endoftext|> violated her probation. **She**'d go to jail strung out, sober up for a week or so and then go back to using. St

Ex 4 <|endoftext|> **has** assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The

Ex 5 <|endoftext|> IRAs **and** Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

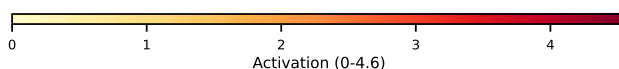
Ex 1 <|endoftext|>~~weary~~ barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the ~~weary~~ spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the ~~weary~~ carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look ~~weary~~ and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a ~~weary~~ Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

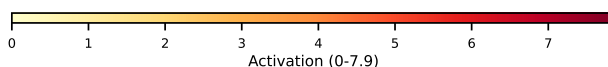
Ex 1 <|endoftext|>Wouldn't the following INCA-descriptor complement the CEFR-descriptors in ~~contents~~ and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|>the ~~contents~~ of the tank. In addition, prolonged contact with tank ~~contents~~, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection ~~contents~~ 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|>those non-findcpa.com.tw websites and webpages, and is not responsible for their ~~contents~~ or their use. By linking to a non

Ex 5 , and copy all the ~~contents~~. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

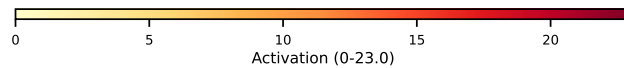


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarner, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

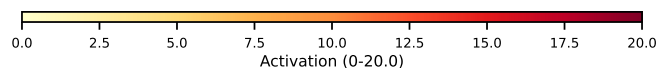


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



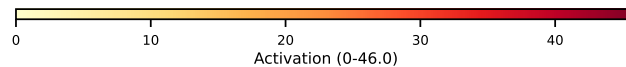
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Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

- Ex 1** ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year
- Ex 2** random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6
- Ex 3** a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent
- Ex 4** generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University
- Ex 5** radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects

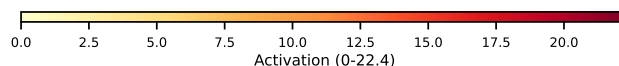


Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

- Ex 1** <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more
- Ex 2** . "Spurred by the growing body of scientific research showing the broad harm ne**onic**otinoids pose to bees and other pollinators, earlier this year
- Ex 3** in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.
- Ex 4** human-made pesticides, ne**onic**otinoids, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm
- Ex 5** , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

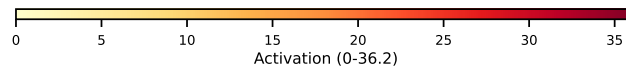
Ex 1 ,450 in the medium-term. On the other^{hand}, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other^{hand}, as a result of

Ex 3 ezers, a steady scott^{hand}. eAuditNet specifications is web-based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue-filled syringes, cast, fabricated part is^{hand}- scott assembled using two

Ex 5 On the other^{hand}, the price of the base version of the JAWA Forty Two starts at 000 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

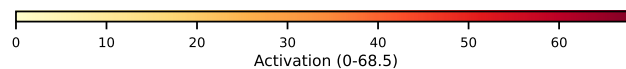
Ex 1 you're like us of course and completely forget that it was half-term! Unda^{unted}, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-val^{unted} Google Glasses. Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh^{unted} by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha^{unted} Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue unda^{unted} by the task in hand. Our thoughts and prayers are now with his family,



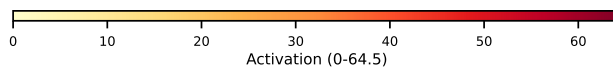
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Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained **record** of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial **record** of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a **record** 1.2 million bpd
- Ex 4** ain a concise **record** inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping **record** the

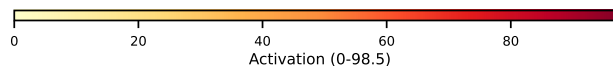


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his **blind** along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to **blind** study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of **blind** marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** **blind**. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally **blind**. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual



Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

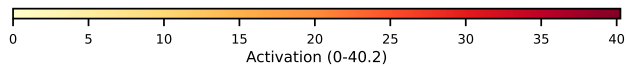
Ex 1 PCDR to thwart BPF rule in BTC

Ex 2 uvo with faster loading, FPS and frame-**times**. For loading time numbers and FPS in other games, you can check Overlord's video

Ex 3 was originally published on Assam Times.Michael Jackson's *Great Beer Guide

Ex 4 primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for story**times**." Written by Lynn Moroney

Ex 5 also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-**times** show a 16 ms minimum and



Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

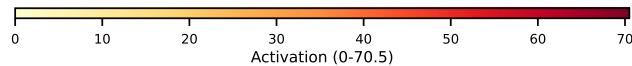
Ex 1 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: **Pres**erving the integrity of original records...including the mistakes! As primary

Ex 2 79 www.pcmc.co.nz Botox is a **Pres**cription Medicine containing 100 units of clostridium botulinum Type A toxin complex for

Ex 3 **Pres**umably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously

Ex 4 available from The **Pres**erve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a

Ex 5 of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five **Pres**idents'

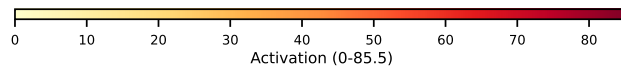


Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she **looks** in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now **looks** to be resorting to cronyism. This will not
- Ex 3** a form of despotism that **looks** like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and **looks** convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading **looks** a lot like cultivating a garden. Man, what a great

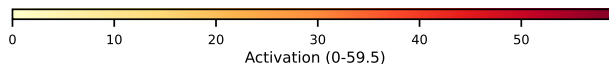


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



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Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

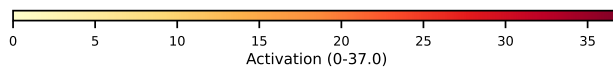
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

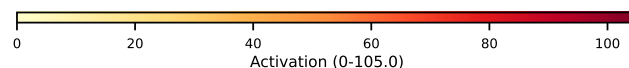
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

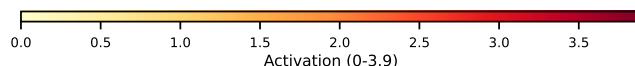
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. Fuel injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - Fueled Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron Suelzle said. Lewis is a cousin to the Suelzle family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (Lycaeides melissa samuelis), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: Dual disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

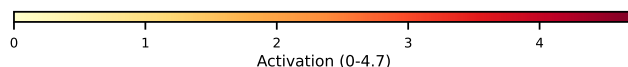
Ex 1 <|endoftext|> to another. The provider of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare provider about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical provider to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new provider, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The provider informed us that they were not able to do the service. (I believe that



Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

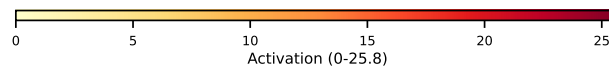
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right sm**ack** dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." **Nic**ola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-**Nicely** in Guys and Dolls

Ex 4 **Cack**ling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecr**ack**ers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

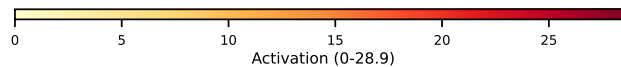
Ex 1 clothing range from **laid** back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm **audio** jack will work with multi-room **audio**. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room **audio** should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room **audio** feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket **audio** systems controlled by near-microscopic buttons. It takes a

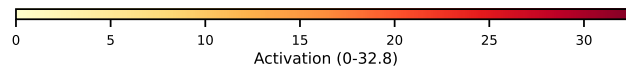


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

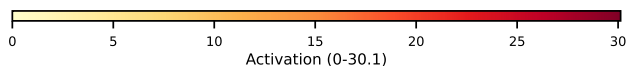


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Re ceiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



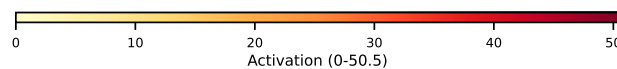
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Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

- Ex 1** NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost
- Ex 2** it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail
- Ex 3** Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare
- Ex 4** stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.
- Ex 5** el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of

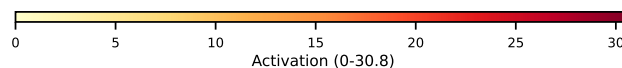


Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

- Ex 1** and literature is also passed down throughgenerations, thus shaping the culture of that community and taking years to form. A country's history has a major role in
- Ex 2** and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Government al Organizations' Network (PNGO
- Ex 3** college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice
- Ex 4** and review of all scientific literature available as well as my own clinicalobservations, I have preparedextensiveguidelines for you to follow during the 7 Stage Fat On The Move
- Ex 5** choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



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Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

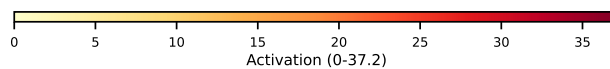
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

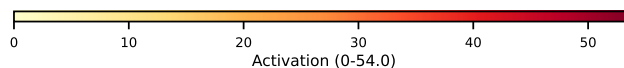
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

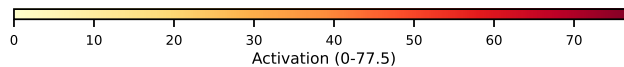
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

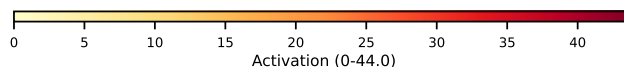
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back-EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

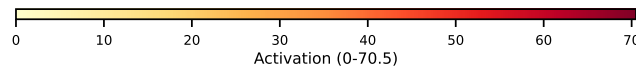
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

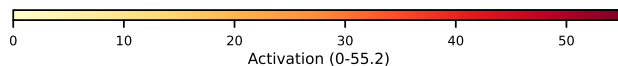
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The

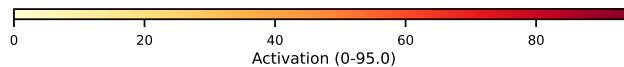


Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said **Ethel** Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** **Ethel**wald of Deira. the Danes and never restored. venerated at Charl bury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters . "I'm pleased to welcome **Ethel** into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors **Ethel** Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.**the**luckyelephant.wordpress

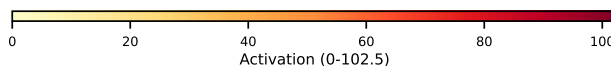


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly **abducted**. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. **Conducted** as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was **abducted** in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and **abducted** Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. **Conducted** retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

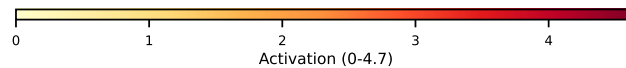
Ex 1 <|endoftext|>Karla **Gutierrez**, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia **Gutierrez** Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain **Gutters** perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, **Gutters** & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond **Gutierrez** and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

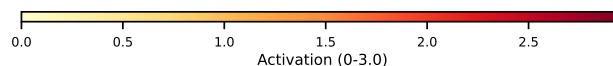
Ex 1 read this article for review of the **rules** or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the **rules** and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old **rules**, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust **rules** governing health insurance companies

Ex 5 **rules** cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to



Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the levers 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, subversive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Covers, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relievers he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet beavers is cute. All of this, by the way, is not meant to

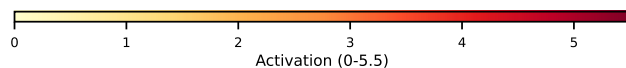


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ARTHROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ARONET, Facebook LA AR
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ARONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ARC, IN
- Ex 5** 0.85. ARRIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru



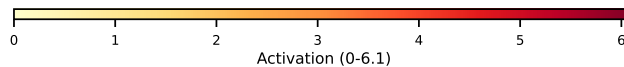
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Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua k utipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

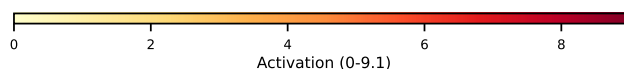


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common king fishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

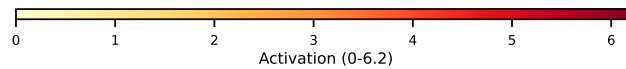
Ex 1 products and ingredients. It has also been confirmed that Lise Wat**ier** products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus**ier** locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did**ier** revised and corrected with the

Ex 4 Made Eas**ier**. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth**ier** than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

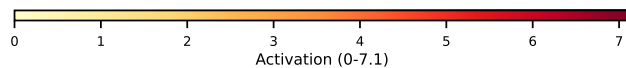
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob**ster** drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham**ster**'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road**ster** carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll**ster** since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang**ster** film -



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Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

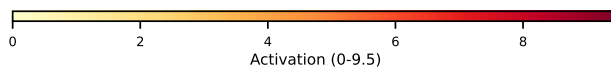
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right
. The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require
three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet
that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a
walk on the beach or climbing a mountain, or going to a national park or
going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it.
One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

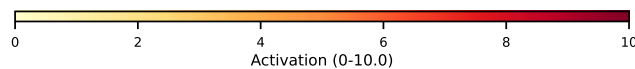
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes
misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede
del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715. Its walking distance
from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium
, Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price
second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

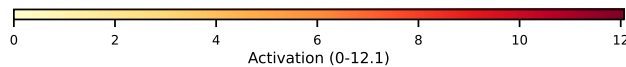
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

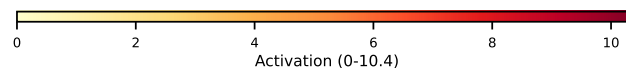
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared

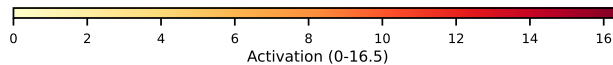


Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

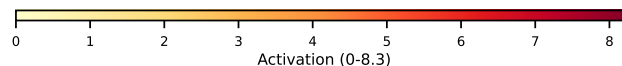


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

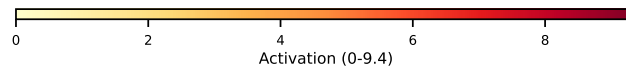


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

